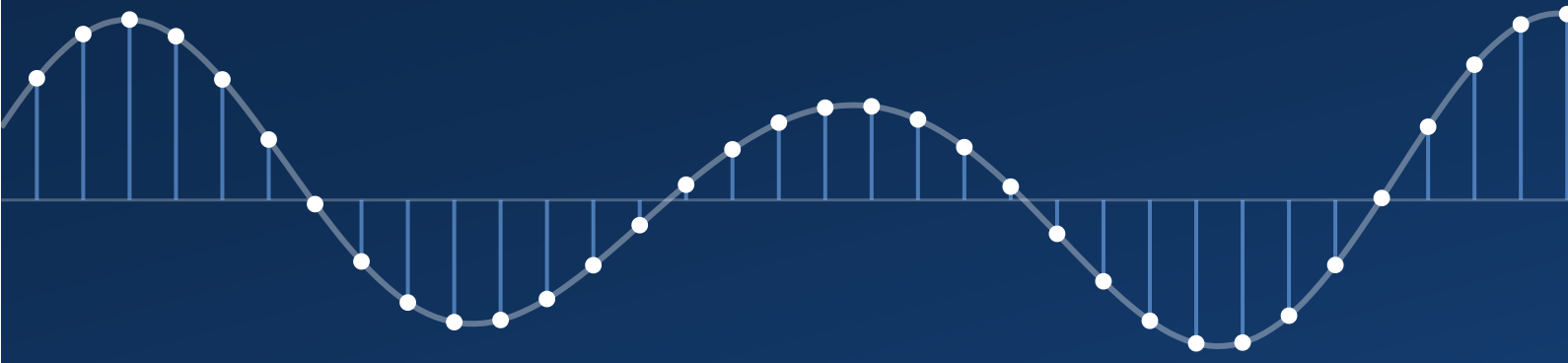
DIGITAL SIGNAL PROCESSING

CHAPTER 02

النظري والمسائل

كل الأسئلة النظري من محاضرات 5 و 6، وكل المسائل من المحاضرات و Sheet 2 والامتحانات القديمة، بإجابات نموذجية بجُمل الدكتور وطريقته في الحل.

DFT · IDFT · Parseval · FFT & Butterfly · IFFT



13	12	+20
سؤال نظري	مسألة محلولة	امتحان قديم متراجع

المحتوى

01 **محاضرة 5**
DFT, IDFT & Properties
8 سؤال Q1 – Q8

02 **محاضرة 6**
FFT, Butterfly & IFFT
5 سؤال Q9 – Q13

03 **مسائل المحاضرات**
Examples 2.1 لحد 2.4
4 سؤال P1 – P4

04 **مسائل Sheet 2**
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05 **مسائل الامتحانات القديمة**
4 سؤال P9 – P12

إزاي تستخدم الملف

النظري بجُمل الدكتور من محاضرات 5 و 6، مترتب (تعريف ← شرح ← معادلة ← رسمة) عشان يبان إنك فاهم. والمسائل خطوة بخطوة بنفس طريقة الدكتور: الـ DFT زي Example 2.1، والـ IDFT زي Example 2.2، والـ FFT زي Example 2.3 و Fig. 2.3، والـ IFFT بنفس الـ flowgraph زي Sec. 2.6. والسطر البرتقالي فيه الامتحانات اللي السؤال جه فيها قبل كده.

★ الأسئلة اللي بتتكرر أكتر (ابدأ بيها)

السؤال	عدد مرات وروده
Q3. Prove that the DFT, X(k), of a sequence x(n) is periodic.	13
P9. x(n) = {1, 0, 1, 1} — DFT / FFT / IFFT / Parseval / spectra / saving	10
P10. x(n) = {1, 0, 1, 0} — DFT / FFT / IFFT / Parseval / spectra / saving	4
P12. x(n) = {1, 0, 0, 1} — DFT / FFT / IFFT / Parseval / spectra / saving	3
Q8. Check (✓) or (✗) and correct the mistakes (DFT / FT statements).	3

SECTION 01

محاضرة 5

DFT, IDFT & Properties

1. Define the Discrete Fourier Transform (DFT).
2. Find the relation between the DFT and the Fourier transform.
3. Prove that the DFT, X(k), of a sequence x(n) is periodic.
4. Show that the amplitude spectrum of an N-point DFT is symmetrical about harmonic N/2 (folding f...
5. Define the Inverse Discrete Fourier Transform (IDFT).
6. State the mathematical properties of the DFT.
7. State Parseval's theorem.
8. Check (✓) or (✗) and correct the mistakes (DFT / FT statements).

Q1 Define the Discrete Fourier Transform (DFT).

ANSWER

Assume a waveform has been sampled at regular time intervals T to produce the sample sequence {x(nT)} = x(0), x(T), …, x[(N − 1)T] of N sample values. The DFT of x(nT) is defined as the sequence of complex values {X(kΩ)} = X(0), X(Ω), …, X[(N − 1)Ω] in the frequency domain, where Ω = 2π/NT is the first harmonic frequency:

$$X(k) = \Sigma_{n=0}^{N-1} x(nT)\, e^{-jk\Omega nT}, \quad k = 0, 1, \ldots, N-1 \quad (2.4)$$

$$X(k) = R(k) + jI(k) \quad (2.1) \qquad |X(k)| = [R^2(k) + I^2(k)]^{1/2} \quad (2.2) \qquad \Phi(k) = \tan^{-1}[I(k)/R(k)] \quad (2.3)$$

Book exam Midterm 2015 جه في

Q2 Find the relation between the DFT and the Fourier transform.

ANSWER

Equation (2.4) is analogous to the Fourier transform $F(j\omega) = \int f(t)\, e^{-j\omega t}\, dt$ (F-1). Putting x(nT) = f(t), kΩ = ω, nT = t, dt = T, and replacing the integral by a summation (for f(t) = 0 outside 0 ≤ t ≤ (N − 1)T):

$$\Sigma_{n=0}^{N-1} x(nT)\, e^{-jk\Omega nT}\, T = F(j\omega) \quad (2.5)$$

$$\mathbf{F(j\omega) = X(k)\, T} \quad \text{i.e.} \quad \mathbf{FT = DFT \times T} \quad (2.6)$$

The FT components are obtained by multiplying the DFT components by the sampling interval T. Similarly, **IFT = IDFT / T**.

Q3 Prove that the DFT, X(k), of a sequence x(n) is periodic.

متكرر في الامتحانات

ANSWER

$$X(k) = \Sigma_{n=0}^{N-1} x(nT)\, e^{-j2\pi kn/N}$$

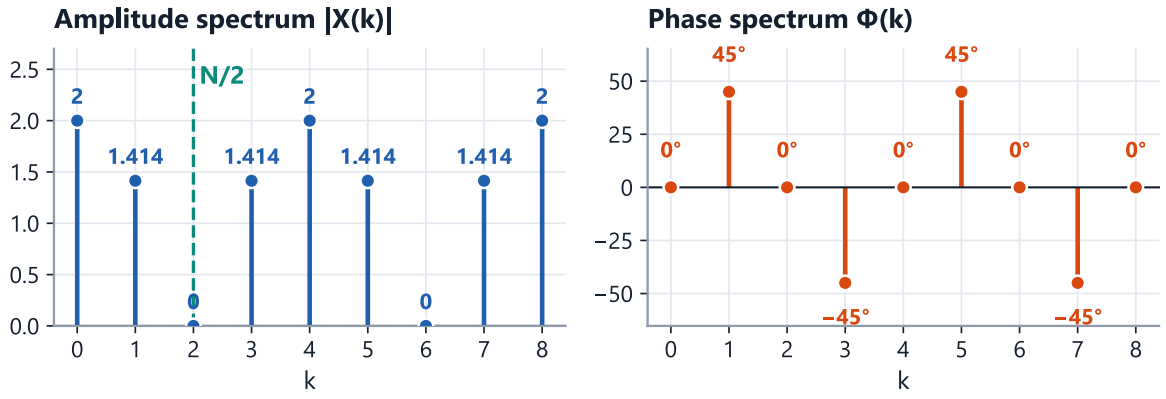
$$X(k+N) = \Sigma_{n=0}^{N-1} x(nT)\, e^{-j2\pi kn/N} \cdot e^{-j2\pi Nn/N} = \Sigma_{n=0}^{N-1} x(nT)\, e^{-j2\pi kn/N} \cdot e^{-j2\pi n}$$

Since n is integral, $e^{-j2\pi n} = 1$, so:

$$X(k + N) = X(k)$$

The DFT is periodic with period N (the cyclical property of the DFT). E.g. for {1, 0, 0, 1}: X(0) = 2, therefore X(4) = 2 also.

|X(k)| for {1, 0, 0, 1}: X(k + 4) = X(k), symmetric about N/2



Final 23-1-2023 (Q3-a) Final 17-8-2026 (Q2-a) Final 9-6-2026 (Q2-a) Final 7-6-2023 (Q1-b) جه في

Midterm 12-11-2025 (Q2-a) Final 23-12-2017 (Q2-b) Final 2-1-2017 (Q2-b) Final 12-1-2019 (Q2-b)

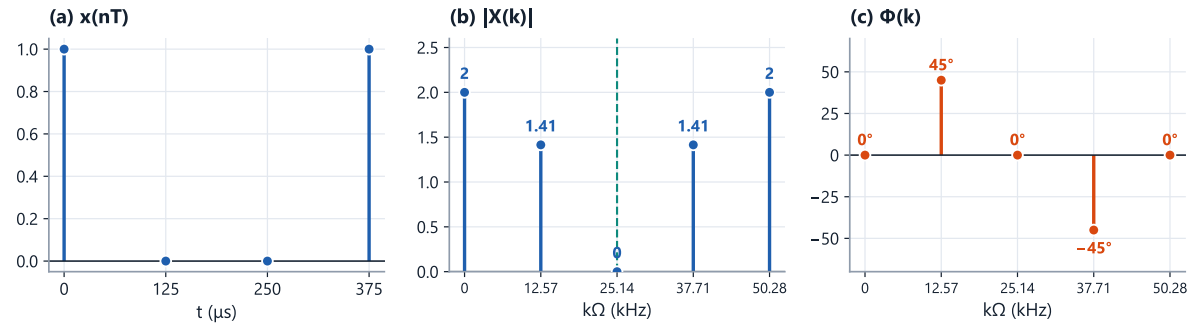
Midterm 2 (Q2-b) Midterm 1 Model-2 (Q2-a) Midterm 8-12-2020 (Q2-b) Midterm 28-11-2021 (Q2-a)

Sheet 2 (2.1)

Q4 Show that the amplitude spectrum of an N-point DFT is symmetrical about harmonic N/2 (folding frequency).

ANSWER

- The amplitude spectrum of an N-point DFT is symmetrical about harmonic N/2, and the phase angles are an odd function centred round this component.
- If $2f_{max}$ samples per second are taken for t seconds, then $2f_{max}t = N$, and the first harmonic frequency is $1/t = 2f_{max}/N$.
- The symmetry at harmonic N/2 occurs at the frequency $(N/2)(2f_{max}/N) = \mathbf{f_{max}}$, the maximum frequency present in the signal (the **folding frequency**).
- So it is unnecessary to plot beyond N/2: N real data values transform to N/2 complex DFT values of practical significance.



Q5 Define the Inverse Discrete Fourier Transform (IDFT).

ANSWER

To obtain the discrete transformation from the frequency to the time domain, the IDFT is used:

$$x(nT) = F_D^{-1}[X(k)] = (1/N) \sum_{k=0}^{N-1} X(k)\, e^{jk\Omega nT}, \quad n = 0, 1, \ldots, N-1 \quad (2.7)$$

The inverse Fourier transform is obtained from the IDFT by dividing by T: **IFT = IDFT / T**.

Q6 State the mathematical properties of the DFT.

ANSWER

1. **Symmetry:** $\mathrm{Re}[X(N-k)] = \mathrm{Re}\, X(k)$ (2.8) and $\mathrm{Im}[X(N-k)] = -\mathrm{Im}[X(k)]$ (2.9).
2. **Even functions:** if $x_e(n) = x_e(-n)$: $X_e(k) = \Sigma\, x_e(n) \cos(k\Omega nT)$ (2.10).
3. **Odd functions:** if $x_o(n) = -x_o(-n)$: $X_o(k) = -j\, \Sigma\, x_o(n) \sin(k\Omega nT)$ (2.11).
4. **Parseval's theorem:** $\Sigma\, x^2(n) = (1/N)\, \Sigma\, |X(k)|^2$ (2.12).
5. **Delta function:** $F_D[\delta(nT)] = 1$ (2.13).
6. Linear cross-correlation of two sequences may be computed using DFTs.
7. DFTs may be used to compute circular convolutions and, using augmenting zeros, linear convolutions.

Final 18-1-2024 (Q2-a) جه في

Q7 State Parseval's theorem.

ANSWER

The normalized energy in the signal is given by either of the expressions:

$$\sum_{n=0}^{N-1} x^2(n) = (1/N) \sum_{k=0}^{N-1} |X(k)|^2 \quad (2.12)$$

The left-hand side is the sum of the squared magnitudes of the time series; the right-hand side is the mean square spectral amplitude.

Q8 Check (✓) or (✗) and correct the mistakes (DFT / FT statements).

ANSWER

1. Fourier transform (FT) is calculated for non-periodic continuous-time signals. **(✓)**
2. Discrete Fourier transform (DFT) is obtained for the analog signals. **(✗)**
   Correct: the DFT is obtained for the sampled (discrete-time) sequence x(nT).

Quiz 1 Summer 26 (Q3) Final 23-12-2017 (Q5-b) Final 12-1-2019 (Q5) جه في

SECTION 02

محاضرة 6

FFT, Butterfly & IFFT

1. Discuss the computational complexity of the DFT.

2. Explain the decimation-in-time (DIT) FFT algorithm.

3. What is the butterfly? Draw it.

4. What are the computational advantages (saving) of the FFT?

5. Show how the IFFT is obtained from the FFT.

Q9 Discuss the computational complexity of the DFT.

ANSWER

For an 8-point DFT (Eq. 2.15): $X(k) = x(0)e^{-jK0} + x(1)e^{-jK1} + \ldots + x(7)e^{-jK7}$, $k = 0, \ldots, 7$.

- Each X(k) needs 8 complex multiplications and 7 complex additions; for N points: N and N − 1.
- There are N harmonic components to evaluate (k = 0 … N − 1).

Operation	DFT
No. of complex multiplications	N^2
No. of complex additions	N(N − 1)

If N = 1024, approximately one million complex multiplications and one million complex additions are required, so some means of reducing these numbers is desirable.

Q10 Explain the decimation-in-time (DIT) FFT algorithm.

ANSWER

- The FFT uses the computational redundancy in the DFT to reduce the number of calculations (Cooley and Tukey, 1965). Decimation refers to the reduction in calculations performed on time-domain data.
- Writing $W_N = e^{-j2\pi/N}$ (2.17), the DFT becomes $X(k) = \Sigma\, x_n\, W_N^{kn}$ (2.18), with:

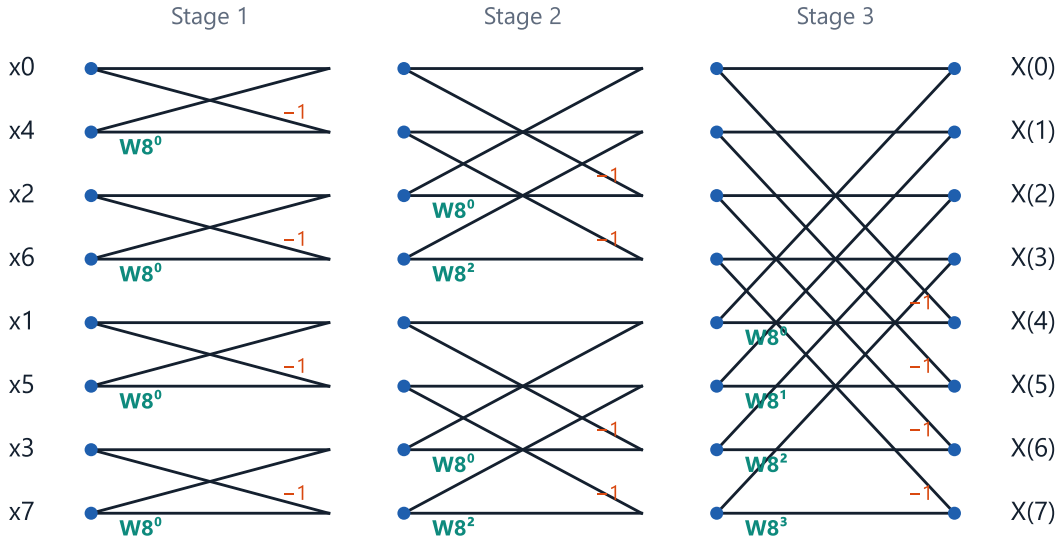
$$W_N^2 = W_{N/2} \text{ (2.21b)} \qquad W_N^{k+N/2} = -W_N^k \text{ (2.21c)}$$

- The data are split into the even sequence x_{2n} and the odd sequence x_{2n+1} (N/2 points each):

$$X1(k) = X11(k) + W_N^k\, X12(k) \text{ (2.24)}$$

- X11(k) is the DFT of the even sequence and X12(k) of the odd sequence. Repeating the decomposition, an 8-point DFT becomes four 2-point DFTs (Table 2.1).

8-point DIT FFT: N/2 butterflies × $\log_2 N$ stages (Fig. 2.3)



Q11 What is the butterfly? Draw it.

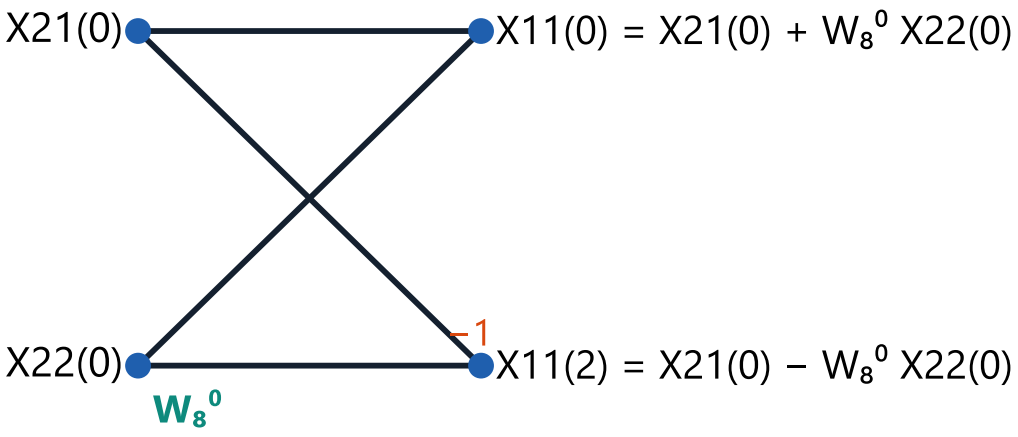
ANSWER

Equations 2.25 are represented diagrammatically by taking them in pairs, exploiting the symmetry centred on the sign differences:

$$X11(0) = X21(0) + W_8^0\, X22(0) \qquad X11(2) = X21(0) - W_8^0\, X22(0)$$

The inputs are at the left of the cross and the outputs at the right. The structure is called a **butterfly**, being reminiscent of the symbolic representation of the insect (Fig. 2.2).

The butterfly (Fig. 2.2 a)



Q12 What are the computational advantages (saving) of the FFT?

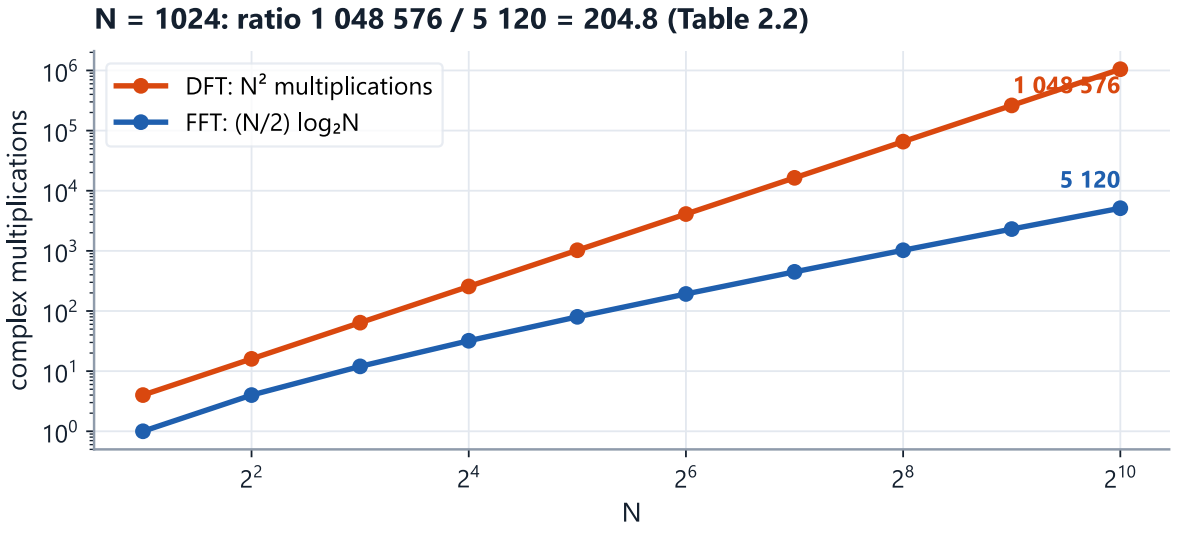
ANSWER

- An N-point FFT contains N/2 butterflies per stage and $\log_2 N$ stages: $(N/2)\log_2 N$ butterflies.
- Each butterfly involves one complex multiplication and two complex additions.

	DFT	FFT	Saving
Complex multiplications	N^2	$(N/2)\log_2 N$	$N^2 - (N/2)\log_2 N$
Complex additions	$N(N-1)$	$N\log_2 N$	$N(N-1) - N\log_2 N$

N	DFT mult.	DFT add.	FFT mult.	FFT add.
4	16	12	4	8
8	64	56	12	24
1024	1 048 576	1 047 552	5 120	10 240

For N = 1024 the number of multiplications is reduced by a factor of 204.8.



Q13 Show how the IFFT is obtained from the FFT.

ANSWER

Substituting in Eq. 2.4 (sum over λ, k → μ, Ω = 2π/NT):

$$\text{FFT:}\quad X(\mu) = \Sigma_{\lambda=0}^{N-1}\, x(\lambda)\, e^{-j\mu(2\pi/N)\lambda} \quad (2.26)$$

and similarly in Eq. 2.7 (k = λ, n = μ):

$$\text{IFFT:}\quad x(\mu) = (1/N)\, \Sigma_{\lambda=0}^{N-1}\, X(\lambda)\, e^{j\mu(2\pi/N)\lambda} \quad (2.27)$$

The IFFT differs from the FFT only in the factor **1/N** and the **sign of the exponent**, so with small modifications the FFT may be used to calculate the IFFT.

Procedure: re-order X(k) as in the FFT, use the same butterflies with W_N^{-k} instead of W_N^{k}, then divide the outputs by N.

SECTION 03

مسائل المحاضرات

Examples 2.1 لحد 2.4

1. Example 2.1 — DFT of {1, 0, 0, 1}
2. Example 2.2 — IDFT of {2, 1 + j, 0, 1 − j}
3. Example 2.3 — FFT of {1, 0, 0, 1}
4. Example 2.4 — FFT of {0.5, 1, 1, 0.5}, sampled at 125 Hz

P1 Example 2.1 — DFT of {1, 0, 0, 1}

The DFT of the sequence {1, 0, 0, 1} (sampled at 8 kHz) will be evaluated. Find X(k), plot |X(k)| and Φ(k), and find the Fourier transform components.

SOLUTION

$$X(k) = \Sigma_{n=0}^{N-1} x(nT)\, e^{-jk\Omega nT} = \Sigma_{n=0}^{N-1} x(nT)\, e^{-j2\pi kn/N}, \quad \Omega = 2\pi/NT, \quad N = 4 \tag{2.4}$$

x(0) = 1, x(T) = 0, x(2T) = 0, x(3T) = 1. It is required to find X(k) for k = 0, 1, 2, 3.

With k = 0:

$$\begin{aligned} X(0) &= \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(0)n/4} \\ &= x(0) + x(T) + x(2T) + x(3T) \\ &= 1 + 0 + 0 + 1 \\ &= \mathbf{2} \end{aligned}$$

Thus X(0) = 2, of magnitude 2 and phase angle Φ(0) = 0°.

With k = 1:

$$\begin{aligned} X(1) &= \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(1)n/4} \\ &= 1 + 0 + 0 + e^{-j3\pi/2} \\ &= 1 + [\cos(3\pi/2) - j\sin(3\pi/2)] \\ &= 1 + j \\ &= \mathbf{1 + j} \end{aligned}$$

Thus X(1) = 1 + j, of magnitude √2 = 1.414 and phase angle Φ(1) = 45°.

With k = 2:

$$\begin{aligned} X(2) &= \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(2)n/4} \\ &= 1 + 0 + 0 + e^{-j3\pi} \\ &= 1 + [\cos(3\pi) - j\sin(3\pi)] \\ &= 1 + (-1) \\ &= \mathbf{0} \end{aligned}$$

Thus X(2) = 0, of magnitude 0 and phase angle Φ(2) = 0°.

With k = 3:

$$\begin{aligned} X(3) &= \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(3)n/4} \\ &= 1 + 0 + 0 + e^{-j9\pi/2} \\ &= 1 + [\cos(9\pi/2) - j\sin(9\pi/2)] \end{aligned}$$

$$= 1 + (-j)$$
$$= \mathbf{1 - j}$$

Thus X(3) = 1 − j, of magnitude √2 = 1.414 and phase angle Φ(3) = −45°.

So, the time series x(n) = {1, 0, 0, 1} has the DFT given by **X(k) = {2, 1 + j, 0, 1 − j}**.

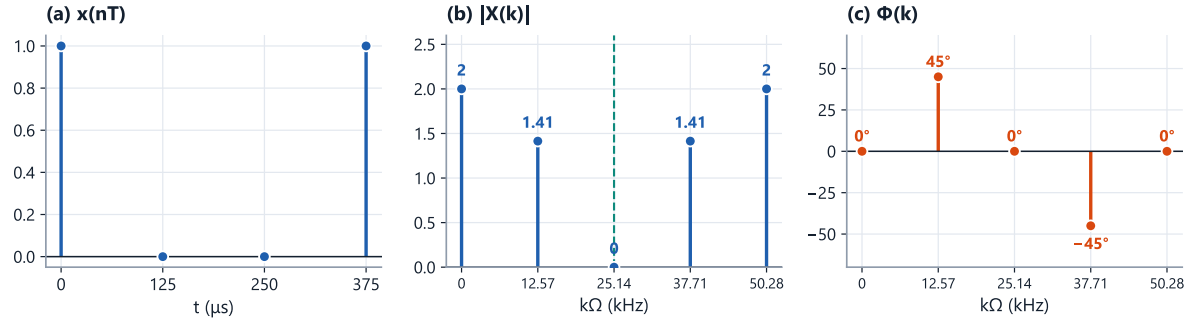
Amplitude and phase spectra:

k	0	1	2	3
X(k)	2	1 + j	0	1 − j
|X(k)|	2	√2 = 1.414	0	√2 = 1.414
Φ(k)	0°	45°	0°	−45°

$$T = 1/(8\times10^{3}) = 125\ \mu s \quad \Omega = 2\pi/NT = 2\pi/(4 \times 125\times10^{-6}) = 12.57\ k \rightarrow 2\Omega = 25.14\ k,\ 3\Omega = 37.71\ k$$

The Fourier transform components are obtained by multiplying the DFT components by T = 125 μs (FT = DFT × T):

$$F(0) = 250\ \mu V\ Hz^{-1}, \quad F(\Omega) = (125 + j125)\ \mu V\ Hz^{-1}, \quad F(2\Omega) = 0, \quad F(3\Omega) = (125 - j125)\ \mu V\ Hz^{-1}$$



P2 Example 2.2 — IDFT of {2, 1 + j, 0, 1 − j}

Derive the time series {1, 0, 0, 1} from its DFT components {2, 1 + j, 0, 1 − j}.

SOLUTION

$$x(nT) = F_D^{-1}[X(k)] = (1/N)\ \Sigma_{k=0}^{N-1}\ X(k)\ e^{jk\Omega nT} = (1/N)\ \Sigma_{k=0}^{N-1}\ X(k)\ e^{j2\pi kn/N}\ (2.7)$$

With n = 0:

$$x(nT) = x(0) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(0)/4}$$
$$= ¼\ [2 + (1 + j) + 0 + (1 - j)]$$

$$= \mathbf{1}, \text{ as expected.}$$

With n = 1:

$$x(nT) = x(T) = (1/N)\,\Sigma_{k=0}^{3}\, X(k)\, e^{j2\pi k(1)/4}$$
$$= \tfrac{1}{4}\,[2 + (1 + j)\, e^{j\pi/2} + 0 + (1 - j)\, e^{j3\pi/2}]$$
$$= \tfrac{1}{4}\,[2 + (1 + j)(j) + (1 - j)(-j)]$$
$$= \tfrac{1}{4}\,[2 + (-1 + j) + (-1 - j)]$$
$$= \mathbf{0}, \text{ as expected.}$$

With n = 2:

$$x(nT) = x(2T) = (1/N)\,\Sigma_{k=0}^{3}\, X(k)\, e^{j2\pi k(2)/4}$$
$$= \tfrac{1}{4}\,[2 + (1 + j)\, e^{j\pi} + 0 + (1 - j)\, e^{j3\pi}]$$
$$= \tfrac{1}{4}\,[2 + (1 + j)(-1) + (1 - j)(-1)]$$
$$= \tfrac{1}{4}\,[2 + (-1 - j) + (-1 + j)]$$
$$= \mathbf{0}, \text{ as expected.}$$

With n = 3:

$$x(nT) = x(3T) = (1/N)\,\Sigma_{k=0}^{3}\, X(k)\, e^{j2\pi k(3)/4}$$
$$= \tfrac{1}{4}\,[2 + (1 + j)\, e^{j3\pi/2} + 0 + (1 - j)\, e^{j9\pi/2}]$$
$$= \tfrac{1}{4}\,[2 + (1 + j)(-j) + (1 - j)(j)]$$
$$= \tfrac{1}{4}\,[2 + (1 - j) + (1 + j)]$$
$$= \mathbf{1}, \text{ as expected.}$$

$\therefore$ x(n) = **{1, 0, 0, 1}**, the original sequence.

P3 Example 2.3 — FFT of {1, 0, 0, 1}

Obtain the DFT of {1, 0, 0, 1} by means of the decimation-in-time FFT algorithm.

SOLUTION

This is a four-point DFT with x0 = 1, x1 = 0, x2 = 0, x3 = 1, and

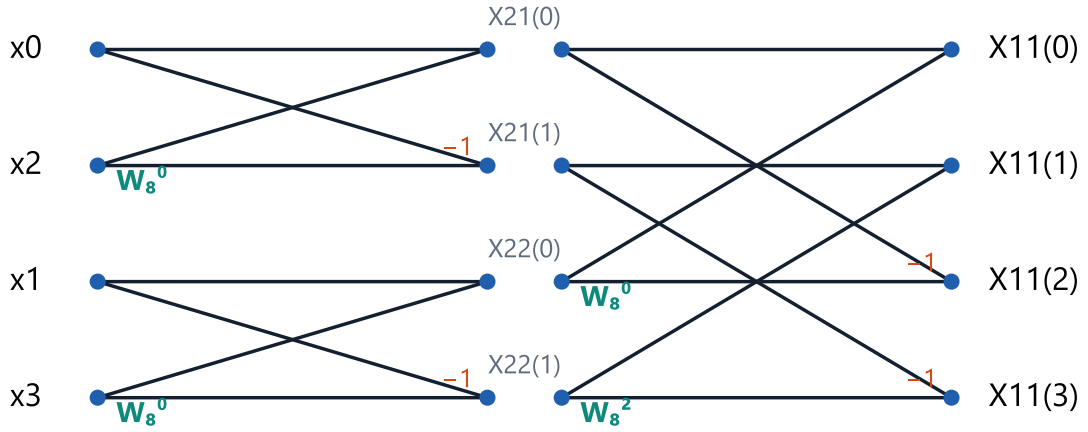
$$X1(k) = X11(k) + W_N^{k}\, X12(k), \quad k = 0, 1, 2, 3 \quad (2.24)$$

The re-ordered sequence is x0, x2, x1, x3. Using the top left-hand corner of Fig. 2.3 ($W_8^{0} = 1$, $W_8^{2} = e^{-j\pi/2} = -j$):

$$X21(0) = x0 + x2 = 1 + 0 = 1$$
$$X21(1) = x0 - x2 = 1 - 0 = 1$$
$$X22(0) = x1 + x3 = 0 + 1 = 1$$
$$X22(1) = x1 - x3 = 0 - 1 = -1$$

$$X11(0) = X21(0) + W_8^0\, X22(0) = 1 + 1 = \mathbf{2}$$
$$X11(1) = X21(1) + W_8^2\, X22(1) = 1 + e^{-j\pi/2}(-1) = 1 - j(-1) = \mathbf{1 + j}$$
$$X11(2) = X21(0) - W_8^0\, X22(0) = 1 - 1 = \mathbf{0}$$
$$X11(3) = X21(1) - W_8^2\, X22(1) = 1 - e^{-j\pi/2}(-1) = 1 + j(-1) = \mathbf{1 - j}$$

4-point DIT FFT (top left-hand corner of Fig. 2.3)



These values are the same as those obtained in Section 2.1, but were more readily obtained using the FFT algorithm.

P4 Example 2.4 — FFT of {0.5, 1, 1, 0.5}, sampled at 125 Hz

The first four sampled voltage values of a 10 Hz bandwidth signal sampled at 125 Hz were (0.5, 1, 1, 0.5). Obtain the DFT using the FFT, and hence the Fourier transform.

SOLUTION

Re-ordered sequence: x(0), x(2), x(1), x(3). $W_4^0 = 1$, $W_4^1 = e^{-j2\pi/4} = -j$

$$G(0) = x(0) + x(2) = 1.5 \quad G(1) = x(0) - x(2) = -0.5$$
$$H(0) = x(1) + x(3) = 1.5 \quad H(1) = x(1) - x(3) = 0.5$$

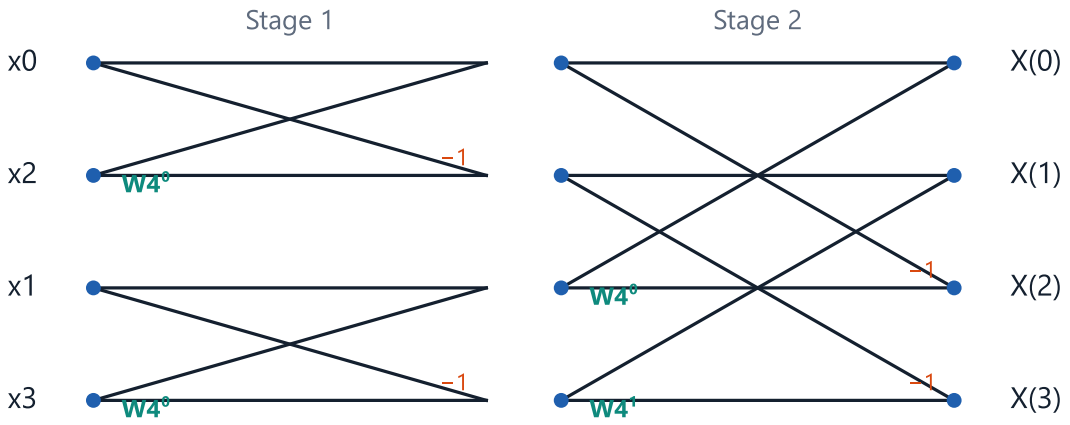
$$X(0) = G(0) + W_4^0 H(0) = 1.5 + 1.5 = \mathbf{3}$$
$$X(1) = G(1) + W_4^1 H(1) = -0.5 - j(0.5) = \mathbf{-0.5 - j0.5}$$

$$X(2) = G(0) - W_4^0 H(0) = 1.5 - 1.5 = \mathbf{0}$$

$$X(3) = G(1) - W_4^1 H(1) = -0.5 + j(0.5) = \mathbf{-0.5 + j0.5}$$

∴ X(k) = **{3, −0.5 − j0.5, 0, −0.5 + j0.5}** (same as the DFT).

4-point DIT FFT flowgraph (Fig. 2.4)



T = 1/125 = 0.008 s; the signal period is 1/10 = 0.1 s → T/period = 0.08 ≪ 1 → FT = T × DFT

FT = {0.024, −0.004(1 + j), 0, 0.004(−1 + j)}

SECTION 04

Sheet 2 مسائل

1. Sheet 2 (2.2 – 2.4) — {0, 1, 1, 0}: DFT, IDFT, Parseval, FFT, saving
2. Sheet 2 (2.5) — 8-point FFT of {0, 0, 1, 1, 1, 1, 0, 0}
3. Sheet 2 (2.6) — x(n) = cos(nπ/2), 8 samples
4. Assignment A2.1 — FFT of {1, 0, 1, 0, 1, 0, 1, 0}

P5 Sheet 2 (2.2 – 2.4) — {0, 1, 1, 0}: DFT, IDFT, Parseval, FFT, saving

2.2 Calculate the DFT of {0, 1, 1, 0} and check by calculating its IDFT.
2.3 Verify Parseval's theorem.
2.4 Calculate it using the DIT FFT, compare numbers of complex additions and multiplications, and calculate the IFFT.

SOLUTION

2.2 DFT (as Example 2.1):

$$X(k) = \Sigma_{n=0}^{N-1} x(nT)\, e^{-jk\Omega nT} = \Sigma_{n=0}^{N-1} x(nT)\, e^{-j2\pi kn/N}, \quad \Omega = 2\pi/NT, \quad N = 4 \ (2.4)$$

x(0) = 0, x(T) = 1, x(2T) = 1, x(3T) = 0. It is required to find X(k) for k = 0, 1, 2, 3.

With k = 0:

$$X(0) = \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(0)n/4}$$
$$= x(0) + x(T) + x(2T) + x(3T)$$
$$= 0 + 1 + 1 + 0$$
$$= \mathbf{2}$$

Thus X(0) = 2, of magnitude 2 and phase angle Φ(0) = 0°.

With k = 1:

$$X(1) = \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(1)n/4}$$
$$= 0 + e^{-j\pi/2} + e^{-j\pi} + 0$$
$$= [\cos(\pi/2) - j\sin(\pi/2)] + [\cos(\pi) - j\sin(\pi)]$$
$$= (-j) + (-1)$$
$$= \mathbf{-1 - j}$$

Thus X(1) = −1 − j, of magnitude √2 = 1.414 and phase angle Φ(1) = −135°.

With k = 2:

$$X(2) = \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(2)n/4}$$
$$= 0 + e^{-j\pi} + e^{-j2\pi} + 0$$
$$= [\cos(\pi) - j\sin(\pi)] + [\cos(2\pi) - j\sin(2\pi)]$$
$$= (-1) + 1$$
$$= \mathbf{0}$$

Thus X(2) = 0, of magnitude 0 and phase angle Φ(2) = 0°.

With k = 3:

$$X(3) = \Sigma_{n=0}^{3}\, x(nT)\, e^{-j2\pi(3)n/4}$$
$$= 0 + e^{-j3\pi/2} + e^{-j3\pi} + 0$$
$$= [\cos(3\pi/2) - j\sin(3\pi/2)] + [\cos(3\pi) - j\sin(3\pi)]$$
$$= j + (-1)$$
$$= \mathbf{-1 + j}$$

Thus X(3) = −1 + j, of magnitude √2 = 1.414 and phase angle Φ(3) = 135°.

So, the time series x(n) = {0, 1, 1, 0} has the DFT given by **X(k) = {2, −1 − j, 0, −1 + j}**.

IDFT check (as Example 2.2):

$$x(nT) = F_D^{-1}[X(k)] = (1/N)\,\Sigma_{k=0}^{N-1}\, X(k)\, e^{jk\Omega nT} = (1/N)\,\Sigma_{k=0}^{N-1}\, X(k)\, e^{j2\pi kn/N} \quad (2.7)$$

With n = 0:

$$x(nT) = x(0) = (1/N)\,\Sigma_{k=0}^{3}\, X(k)\, e^{j2\pi k(0)/4}$$
$$= ¼\,[2 + (-1 - j) + 0 + (-1 + j)]$$
$$= \mathbf{0}, \text{ as expected.}$$

With n = 1:

$$x(nT) = x(T) = (1/N)\,\Sigma_{k=0}^{3}\, X(k)\, e^{j2\pi k(1)/4}$$
$$= ¼\,[2 + (-1 - j)\, e^{j\pi/2} + 0 + (-1 + j)\, e^{j3\pi/2}]$$
$$= ¼\,[2 + (-1 - j)(j) + (-1 + j)(-j)]$$
$$= ¼\,[2 + (1 - j) + (1 + j)]$$
$$= \mathbf{1}, \text{ as expected.}$$

With n = 2:

$$x(nT) = x(2T) = (1/N)\,\Sigma_{k=0}^{3}\, X(k)\, e^{j2\pi k(2)/4}$$
$$= ¼\,[2 + (-1 - j)\, e^{j\pi} + 0 + (-1 + j)\, e^{j3\pi}]$$
$$= ¼\,[2 + (-1 - j)(-1) + (-1 + j)(-1)]$$
$$= ¼\,[2 + (1 + j) + (1 - j)]$$
$$= \mathbf{1}, \text{ as expected.}$$

With n = 3:

$$x(nT) = x(3T) = (1/N)\,\Sigma_{k=0}^{3}\, X(k)\, e^{j2\pi k(3)/4}$$
$$= ¼\,[2 + (-1 - j)\, e^{j3\pi/2} + 0 + (-1 + j)\, e^{j9\pi/2}]$$
$$= ¼\,[2 + (-1 - j)(-j) + (-1 + j)(j)]$$

$$= \tfrac{1}{4}\,[2 + (-1 + j) + (-1 - j)]$$
$$= \mathbf{0}, \text{ as expected.}$$

∴ x(n) = **{0, 1, 1, 0}**, the original sequence.

2.3 Parseval:

$$\Sigma_{n=0}^{N-1}\, x^2(n) = (1/N)\, \Sigma_{k=0}^{N-1}\, |X(k)|^2 \quad (2.12)$$

|X(k)| = {2, √2, 0, √2}

$$\text{L.H.S.} = 0^2 + 1^2 + 1^2 + 0^2 = \mathbf{2}$$
$$\text{R.H.S.} = (1/4)[2^2 + \sqrt{2}^2 + 0^2 + \sqrt{2}^2] = (1/4)(8) = \mathbf{2}$$

L.H.S. = R.H.S., so Parseval's theorem is verified.

2.4 FFT (as Example 2.3):

FFT (as Example 2.3):

This is a four-point DFT with x0 = 0, x1 = 1, x2 = 1, x3 = 0, and

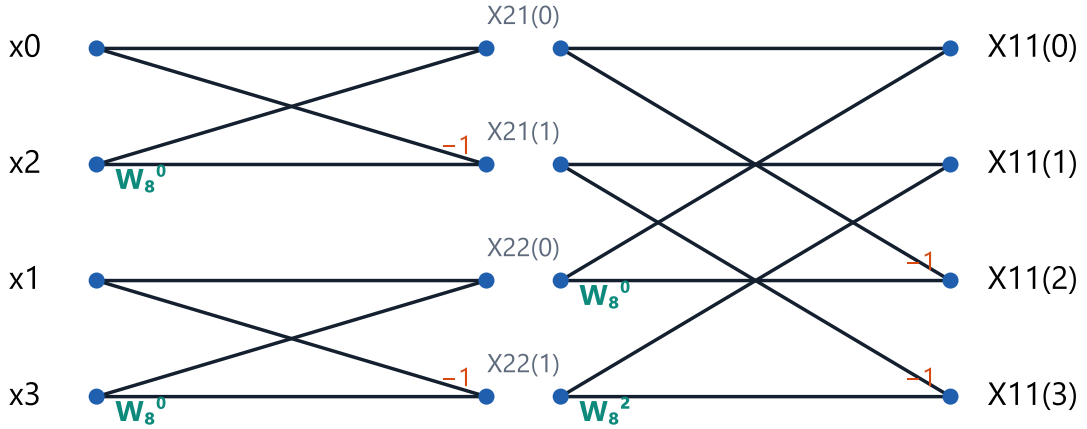
$$X1(k) = X11(k) + W_N^{\,k}\, X12(k), \quad k = 0, 1, 2, 3 \quad (2.24)$$

The re-ordered sequence is x0, x2, x1, x3. Using the top left-hand corner of Fig. 2.3 ($W_8^{\,0} = 1$, $W_8^{\,2} = e^{-j\pi/2} = -j$):

$$X21(0) = x0 + x2 = 0 + 1 = 1$$
$$X21(1) = x0 - x2 = 0 - 1 = -1$$
$$X22(0) = x1 + x3 = 1 + 0 = 1$$
$$X22(1) = x1 - x3 = 1 - 0 = 1$$

$$X11(0) = X21(0) + W_8^{\,0}\, X22(0) = 1 + 1 = \mathbf{2}$$
$$X11(1) = X21(1) + W_8^{\,2}\, X22(1) = -1 + e^{-j\pi/2}(1) = -1 - j(1) = \mathbf{-1 - j}$$
$$X11(2) = X21(0) - W_8^{\,0}\, X22(0) = 1 - 1 = \mathbf{0}$$
$$X11(3) = X21(1) - W_8^{\,2}\, X22(1) = -1 - e^{-j\pi/2}(1) = -1 + j(1) = \mathbf{-1 + j}$$

4-point DIT FFT (top left-hand corner of Fig. 2.3)



These values X(k) = {2, −1 − j, 0, −1 + j} are the same as those obtained in Prob. (2.2), but were more readily obtained using the FFT algorithm.

Comparison (Sec. 2.4, 2.5.2), N = 4:

Operation	DFT	FFT	Saving
No. of complex multiplications	$N^2 = 16$	$(N/2)\log_2 N = 4$	**12**
No. of complex additions	$N(N-1) = 12$	$N\log_2 N = 8$	**4**

IFFT (Sec. 2.6):

The IFFT (2.27) differs from the FFT (2.26) only in the factor 1/N and the sign of the exponent, so the same flowgraph is used with the re-ordered input X(0), X(2), X(1), X(3), $W_8^{-2} = e^{+j\pi/2} = +j$, and the result is divided by N = 4:

$$x21(0) = X(0) + X(2) = 2 \quad x21(1) = X(0) - X(2) = 2$$
$$x22(0) = X(1) + X(3) = -2 \quad x22(1) = X(1) - X(3) = -j2$$

$$x(0) = \tfrac{1}{4}[x21(0) + x22(0)] = \tfrac{1}{4}[0] = \mathbf{0}$$
$$x(1) = \tfrac{1}{4}[x21(1) + j\,x22(1)] = \tfrac{1}{4}[2 + j(-j2)] = \mathbf{1}$$
$$x(2) = \tfrac{1}{4}[x21(0) - x22(0)] = \tfrac{1}{4}[4] = \mathbf{1}$$
$$x(3) = \tfrac{1}{4}[x21(1) - j\,x22(1)] = \tfrac{1}{4}[2 - j(-j2)] = \mathbf{0}$$

∴ x(n) = **{0, 1, 1, 0}**, the original data sequence.

Sheet 2 (2.2, 2.3, 2.4) جه في

P6 Sheet 2 (2.5) — 8-point FFT of {0, 0, 1, 1, 1, 1, 0, 0}

2.5 Calculate the FFT of {0, 0, 1, 1, 1, 1, 0, 0} and plot the amplitude and phase spectra. Check by calculating the IFFT.

SOLUTION

FFT (8-point, Fig. 2.3):

Re-ordered input (Table 2.1): x0, x4, x2, x6, x1, x5, x3, x7. $W_8^0 = 1$, $W_8^1 = (1 - j)/\sqrt{2}$, $W_8^2 = -j$, $W_8^3 = -(1 + j)/\sqrt{2}$

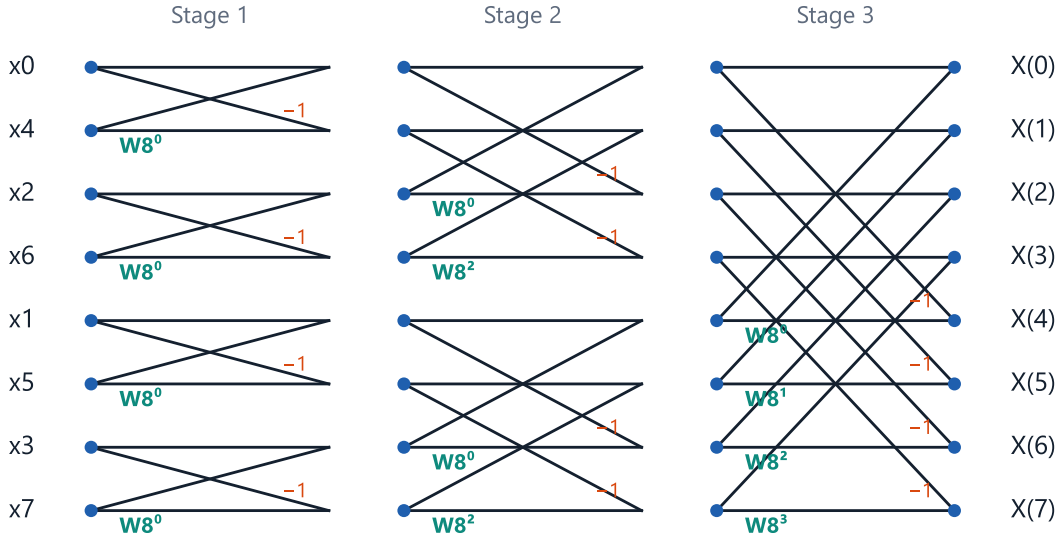
Stage 1: four 2-point DFTs	Stage 2: two 4-point DFTs
$X21(0) = x0 + x4 = 1$ $X21(1) = x0 - x4 = -1$	$X11(0) = X21(0) + W_8^0 X22(0) = 2$ $X11(1) = X21(1) + W_8^2 X22(1) = -1 - j$
$X22(0) = x2 + x6 = 1$ $X22(1) = x2 - x6 = 1$	$X11(2) = X21(0) - W_8^0 X22(0) = 0$ $X11(3) = X21(1) - W_8^2 X22(1) = -1 + j$
$X23(0) = x1 + x5 = 1$ $X23(1) = x1 - x5 = -1$	$X12(0) = X23(0) + W_8^0 X24(0) = 2$ $X12(1) = X23(1) + W_8^2 X24(1) = -1 - j$
$X24(0) = x3 + x7 = 1$ $X24(1) = x3 - x7 = 1$	$X12(2) = X23(0) - W_8^0 X24(0) = 0$ $X12(3) = X23(1) - W_8^2 X24(1) = -1 + j$

Stage 3: $X1(k) = X11(k) + W_8^k X12(k)$, $X1(k + 4) = X11(k) - W_8^k X12(k)$, $k = 0, 1, 2, 3$

$$X(0) = 4 \quad X(4) = 0$$
$$X(1) = -2.414 - j \quad X(5) = 0.414 - j$$
$$X(2) = 0 \quad X(6) = 0$$
$$X(3) = 0.414 + j \quad X(7) = -2.414 + j$$

$\therefore X(k) =$ **{4, −2.414 − j, 0, 0.414 + j, 0, 0.414 − j, 0, −2.414 + j}**

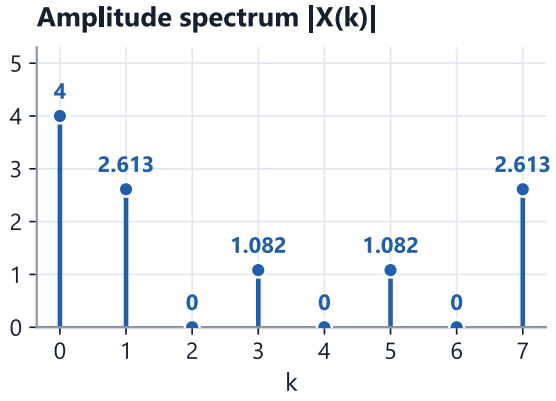
8-point DIT FFT: N/2 butterflies × $\log_2 N$ stages (Fig. 2.3)

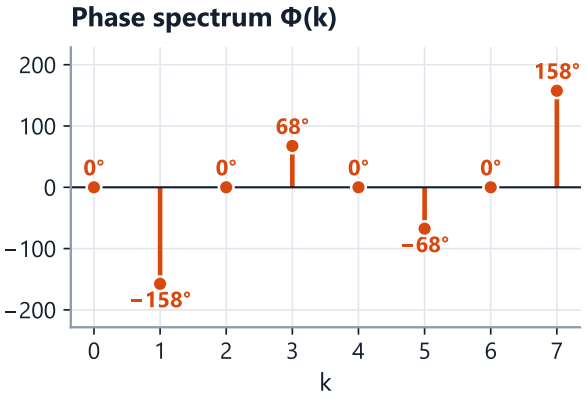


Amplitude and phase spectra (|X(k)| and Φ(k) versus k; symmetrical about N/2 = 4):

k	0	1	2	3	4	5	6	7
X(k)	4	−2.414 − j	0	0.414 + j	0	0.414 − j	0	−2.414 + j
|X(k)|	4	2.613	0	1.082	0	1.082	0	2.613
Φ(k)	0°	−157.5°	0°	67.5°	0°	−67.5°	0°	157.5°

Spectra of {0, 0, 1, 1, 1, 1, 0, 0}





IFFT check (same flowgraph, W_8^{-k}, divided by N = 8):

Re-ordered input (Table 2.1): X(0), X(4), X(2), X(6), X(1), X(5), X(3), X(7). $W_8^0 = 1$, $W_8^{-1} = (1 + j)/\sqrt{2}$, $W_8^{-2} = +j$, $W_8^{-3} = (-1 + j)/\sqrt{2}$

Stage 1: four 2-point DFTs	Stage 2: two 4-point DFTs
X21(0) = X(0) + X(4) = 4 X21(1) = X(0) − X(4) = 4	X11(0) = X21(0) + W_8^0X22(0) = 4 X11(1) = X21(1) + W_8^{-2}X22(1) = 4
X22(0) = X(2) + X(6) = 0 X22(1) = X(2) − X(6) = 0	X11(2) = X21(0) − W_8^0X22(0) = 4 X11(3) = X21(1) − W_8^{-2}X22(1) = 4
X23(0) = X(1) + X(5) = −2 − j2 X23(1) = X(1) − X(5) = −2.828	X12(0) = X23(0) + W_8^0X24(0) = −4 X12(1) = X23(1) + W_8^{-2}X24(1) = −2.828 + j2.828
X24(0) = X(3) + X(7) = −2 + j2 X24(1) = X(3) − X(7) = 2.828	X12(2) = X23(0) − W_8^0X24(0) = −j4 X12(3) = X23(1) − W_8^{-2}X24(1) = −2.828 − j2.828

Stage 3: $X1(k) = X11(k) + W_8^{-k}\, X12(k)$, $X1(k + 4) = X11(k) - W_8^{-k}\, X12(k)$, $k = 0, 1, 2, 3$

x(0) = (1/8)(0) = **0**	x(4) = (1/8)(8) = **1**
x(1) = (1/8)(0) = **0**	x(5) = (1/8)(8) = **1**
x(2) = (1/8)(8) = **1**	x(6) = (1/8)(0) = **0**
x(3) = (1/8)(8) = **1**	x(7) = (1/8)(0) = **0**

∴ x(n) = **{0, 0, 1, 1, 1, 1, 0, 0}**, the original data sequence.

Sheet 2 (2.5) جه في

P7 Sheet 2 (2.6) — x(n) = cos(nπ/2), 8 samples

2.6 For x(n) = cos(nπ/2), find the first 8 samples and then, using the FFT algorithm, calculate X(k). What is the computational saving compared to the DFT?

SOLUTION

$x(n) = \cos(n\pi/2)$: $x(0) = \cos 0 = 1$, $x(1) = \cos(\pi/2) = 0$, $x(2) = \cos \pi = -1$, $x(3) = \cos(3\pi/2) = 0$, …

$x(n) = \mathbf{\{1, 0, -1, 0, 1, 0, -1, 0\}}$

FFT (8-point, Fig. 2.3):

Re-ordered input (Table 2.1): x0, x4, x2, x6, x1, x5, x3, x7. $W_8^0 = 1$, $W_8^1 = (1 - j)/\sqrt{2}$, $W_8^2 = -j$, $W_8^3 = -(1 + j)/\sqrt{2}$

Stage 1: four 2-point DFTs	Stage 2: two 4-point DFTs
X21(0) = x0 + x4 = 2 X21(1) = x0 − x4 = 0	X11(0) = X21(0) + W_8^0X22(0) = 0 X11(1) = X21(1) + W_8^2X22(1) = 0
X22(0) = x2 + x6 = −2 X22(1) = x2 − x6 = 0	X11(2) = X21(0) − W_8^0X22(0) = 4 X11(3) = X21(1) − W_8^2X22(1) = 0
X23(0) = x1 + x5 = 0 X23(1) = x1 − x5 = 0	X12(0) = X23(0) + W_8^0X24(0) = 0 X12(1) = X23(1) + W_8^2X24(1) = 0
X24(0) = x3 + x7 = 0 X24(1) = x3 − x7 = 0	X12(2) = X23(0) − W_8^0X24(0) = 0 X12(3) = X23(1) − W_8^2X24(1) = 0

Stage 3: $X1(k) = X11(k) + W_8^k\, X12(k)$, $X1(k + 4) = X11(k) - W_8^k\, X12(k)$, $k = 0, 1, 2, 3$

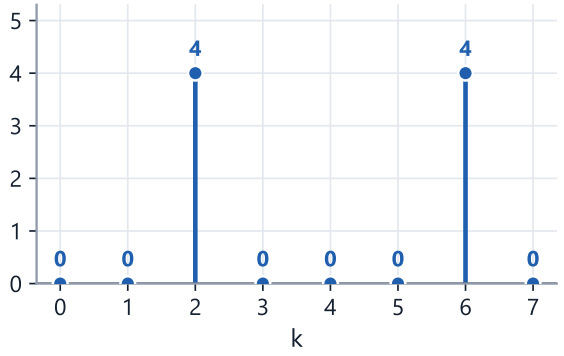
X(0) = 0	X(4) = 0
X(1) = 0	X(5) = 0
X(2) = 4	X(6) = 4
X(3) = 0	X(7) = 0

∴ X(k) = **{0, 0, 4, 0, 0, 0, 4, 0}**

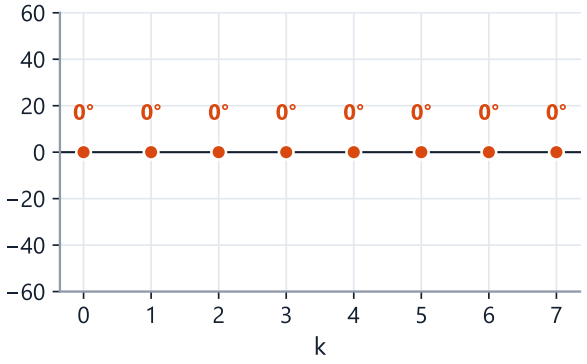
k	0	1	2	3	4	5	6	7
X(k)	0	0	4	0	0	0	4	0
|X(k)|	0	0	4	0	0	0	4	0
Φ(k)	0°	0°	0°	0°	0°	0°	0°	0°

Spectra of x(n) = cos(nπ/2)

Amplitude spectrum |X(k)|



Phase spectrum Φ(k)



Computational saving (Sec. 2.5.2), N = 8:

Operation	DFT	FFT	Saving
No. of complex multiplications	$N^2 = 64$	$(N/2)\log_2 N = 12$	**52**
No. of complex additions	$N(N-1) = 56$	$N\log_2 N = 24$	**32**

Sheet 2 (2.6) جه في

P8 Assignment A2.1 — FFT of {1, 0, 1, 0, 1, 0, 1, 0}

Calculate the FFT of {1, 0, 1, 0, 1, 0, 1, 0}, plot the amplitude and phase spectra and check by the IFFT.

SOLUTION

FFT (8-point, Fig. 2.3):

Re-ordered input (Table 2.1): x0, x4, x2, x6, x1, x5, x3, x7. $W_8^0 = 1$, $W_8^1 = (1-j)/\sqrt{2}$, $W_8^2 = -j$, $W_8^3 = -(1+j)/\sqrt{2}$

Stage 1: four 2-point DFTs	Stage 2: two 4-point DFTs
$X21(0) = x0 + x4 = 2$ $X21(1) = x0 - x4 = 0$	$X11(0) = X21(0) + W_8^0 X22(0) = 4$ $X11(1) = X21(1) + W_8^2 X22(1) = 0$
$X22(0) = x2 + x6 = 2$ $X22(1) = x2 - x6 = 0$	$X11(2) = X21(0) - W_8^0 X22(0) = 0$ $X11(3) = X21(1) - W_8^2 X22(1) = 0$
$X23(0) = x1 + x5 = 0$ $X23(1) = x1 - x5 = 0$	$X12(0) = X23(0) + W_8^0 X24(0) = 0$ $X12(1) = X23(1) + W_8^2 X24(1) = 0$
$X24(0) = x3 + x7 = 0$ $X24(1) = x3 - x7 = 0$	$X12(2) = X23(0) - W_8^0 X24(0) = 0$ $X12(3) = X23(1) - W_8^2 X24(1) = 0$

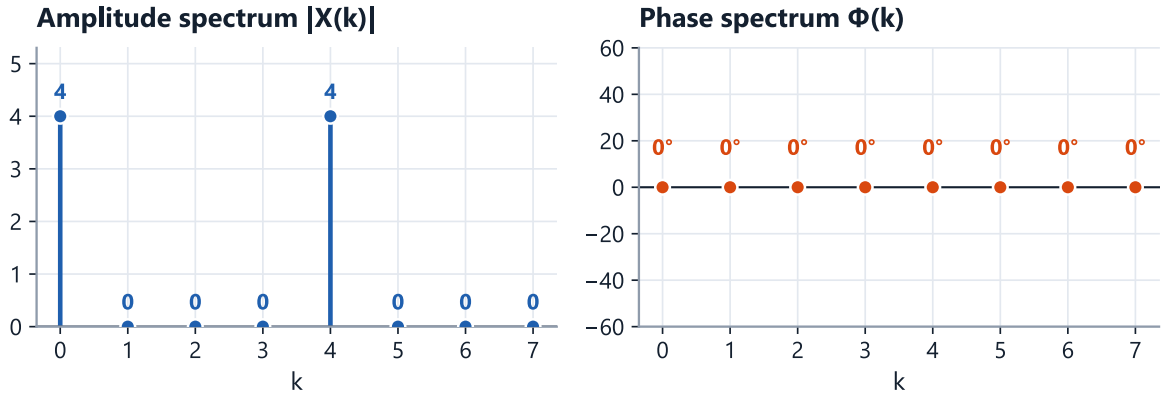
Stage 3: $X1(k) = X11(k) + W_8^k X12(k)$, $X1(k+4) = X11(k) - W_8^k X12(k)$, $k = 0, 1, 2, 3$

$$X(0) = 4 \quad X(4) = 4$$
$$X(1) = 0 \quad X(5) = 0$$
$$X(2) = 0 \quad X(6) = 0$$
$$X(3) = 0 \quad X(7) = 0$$

∴ X(k) = **{4, 0, 0, 0, 4, 0, 0, 0}**

k	0	1	2	3	4	5	6	7
X(k)	4	0	0	0	4	0	0	0
|X(k)|	4	0	0	0	4	0	0	0
Φ(k)	0°	0°	0°	0°	0°	0°	0°	0°

Spectra of {1, 0, 1, 0, 1, 0, 1, 0}



IFFT check:

Re-ordered input (Table 2.1): X(0), X(4), X(2), X(6), X(1), X(5), X(3), X(7). $W_8^{0} = 1$, $W_8^{-1} = (1 + j)/\sqrt{2}$, $W_8^{-2} = +j$, $W_8^{-3} = (-1 + j)/\sqrt{2}$

Stage 1: four 2-point DFTs	Stage 2: two 4-point DFTs
$X21(0) = X(0) + X(4) = 8$ $X21(1) = X(0) - X(4) = 0$	$X11(0) = X21(0) + W_8^{0}X22(0) = 8$ $X11(1) = X21(1) + W_8^{-2}X22(1) = 0$
$X22(0) = X(2) + X(6) = 0$ $X22(1) = X(2) - X(6) = 0$	$X11(2) = X21(0) - W_8^{0}X22(0) = 8$ $X11(3) = X21(1) - W_8^{-2}X22(1) = 0$
$X23(0) = X(1) + X(5) = 0$ $X23(1) = X(1) - X(5) = 0$	$X12(0) = X23(0) + W_8^{0}X24(0) = 0$ $X12(1) = X23(1) + W_8^{-2}X24(1) = 0$
$X24(0) = X(3) + X(7) = 0$ $X24(1) = X(3) - X(7) = 0$	$X12(2) = X23(0) - W_8^{0}X24(0) = 0$ $X12(3) = X23(1) - W_8^{-2}X24(1) = 0$

Stage 3: $X1(k) = X11(k) + W_8^{-k}\, X12(k)$, $X1(k + 4) = X11(k) - W_8^{-k}\, X12(k)$, $k = 0, 1, 2, 3$

$$x(0) = (1/8)(8) = \mathbf{1} \quad x(4) = (1/8)(8) = \mathbf{1}$$
$$x(1) = (1/8)(0) = \mathbf{0} \quad x(5) = (1/8)(0) = \mathbf{0}$$

$$x(2) = (1/8)(8) = \mathbf{1} \qquad x(6) = (1/8)(8) = \mathbf{1}$$
$$x(3) = (1/8)(0) = \mathbf{0} \qquad x(7) = (1/8)(0) = \mathbf{0}$$

∴ x(n) = **{1, 0, 1, 0, 1, 0, 1, 0}**, the original sequence.

Book Assignment 2 (A2.1) **جه في**

SECTION 05

مسائل الامتحانات القديمة

1. x(n) = {1, 0, 1, 1}

2. x(n) = {1, 0, 1, 0}

3. x(n) = {1, 1, 0, 1}

4. x(n) = {1, 0, 0, 1}

P9 x(n) = {1, 0, 1, 1}

متكرر في الامتحانات

For the sequence x(n) = {1, 0, 1, 1}: (i) find the frequency components X(k) using the DFT (or FFT); (ii) check by calculating x(n) using the IFFT (or IDFT); (iii) verify Parseval's theorem; (iv) plot the amplitude and phase spectra; (v) find the computational saving of the FFT.

SOLUTION

(i) DFT method (as Example 2.1):

$$X(k) = \Sigma_{n=0}^{N-1} x(nT)\, e^{-jk\Omega nT} = \Sigma_{n=0}^{N-1} x(nT)\, e^{-j2\pi kn/N}, \quad \Omega = 2\pi/NT, \quad N = 4 \quad (2.4)$$

x(0) = 1, x(T) = 0, x(2T) = 1, x(3T) = 1. It is required to find X(k) for k = 0, 1, 2, 3.

With k = 0:

$$\begin{aligned} X(0) &= \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(0)n/4} \\ &= x(0) + x(T) + x(2T) + x(3T) \\ &= 1 + 0 + 1 + 1 \\ &= \mathbf{3} \end{aligned}$$

Thus X(0) = 3, of magnitude 3 and phase angle Φ(0) = 0°.

With k = 1:

$$\begin{aligned} X(1) &= \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(1)n/4} \\ &= 1 + 0 + e^{-j\pi} + e^{-j3\pi/2} \\ &= 1 + [\cos(\pi) - j\sin(\pi)] + [\cos(3\pi/2) - j\sin(3\pi/2)] \\ &= 1 + (-1) + j \\ &= \mathbf{j} \end{aligned}$$

Thus X(1) = j, of magnitude 1 and phase angle Φ(1) = 90°.

With k = 2:

$$\begin{aligned} X(2) &= \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(2)n/4} \\ &= 1 + 0 + e^{-j2\pi} + e^{-j3\pi} \\ &= 1 + [\cos(2\pi) - j\sin(2\pi)] + [\cos(3\pi) - j\sin(3\pi)] \\ &= 1 + 1 + (-1) \\ &= \mathbf{1} \end{aligned}$$

Thus X(2) = 1, of magnitude 1 and phase angle Φ(2) = 0°.

With k = 3:

$$X(3) = \Sigma_{n=0}^{3}\, x(nT)\, e^{-j2\pi(3)n/4}$$
$$= 1 + 0 + e^{-j3\pi} + e^{-j9\pi/2}$$
$$= 1 + [\cos(3\pi) - j\sin(3\pi)] + [\cos(9\pi/2) - j\sin(9\pi/2)]$$
$$= 1 + (-1) + (-j)$$
$$= \mathbf{-j}$$

Thus X(3) = −j, of magnitude 1 and phase angle Φ(3) = −90°.

So, the time series x(n) = {1, 0, 1, 1} has the DFT given by **X(k) = {3, j, 1, −j}**.

FFT algorithm (as Example 2.3):

This is a four-point DFT with x0 = 1, x1 = 0, x2 = 1, x3 = 1, and

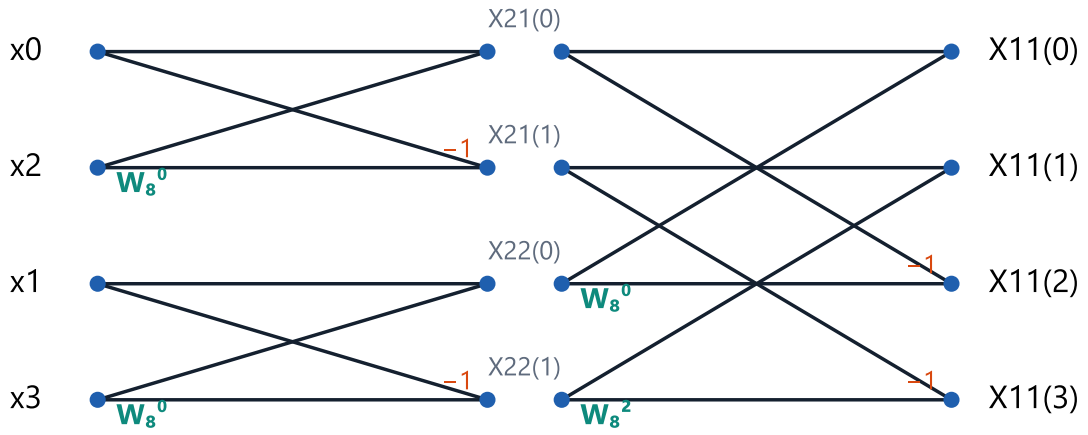
$$X1(k) = X11(k) + W_N^{k}\, X12(k), \quad k = 0, 1, 2, 3 \quad (2.24)$$

The re-ordered sequence is x0, x2, x1, x3. Using the top left-hand corner of Fig. 2.3 ($W_8^0 = 1$, $W_8^2 = e^{-j\pi/2} = -j$):

$$X21(0) = x0 + x2 = 1 + 1 = 2$$
$$X21(1) = x0 - x2 = 1 - 1 = 0$$
$$X22(0) = x1 + x3 = 0 + 1 = 1$$
$$X22(1) = x1 - x3 = 0 - 1 = -1$$

$$X11(0) = X21(0) + W_8^0\, X22(0) = 2 + 1 = \mathbf{3}$$
$$X11(1) = X21(1) + W_8^2\, X22(1) = 0 + e^{-j\pi/2}(-1) = 0 - j(-1) = \mathbf{j}$$
$$X11(2) = X21(0) - W_8^0\, X22(0) = 2 - 1 = \mathbf{1}$$
$$X11(3) = X21(1) - W_8^2\, X22(1) = 0 - e^{-j\pi/2}(-1) = 0 + j(-1) = \mathbf{-j}$$

4-point DIT FFT (top left-hand corner of Fig. 2.3)



These values are the same as those obtained by the DFT.

(ii) Check by the IFFT (Sec. 2.6):

The IFFT (2.27) differs from the FFT (2.26) only in the factor 1/N and the sign of the exponent, so the same flowgraph is used with the re-ordered input X(0), X(2), X(1), X(3), $W_8^{-2} = e^{+j\pi/2} = +j$, and the result is divided by N = 4:

$$x21(0) = X(0) + X(2) = 4 \quad x21(1) = X(0) - X(2) = 2$$
$$x22(0) = X(1) + X(3) = 0 \quad x22(1) = X(1) - X(3) = j2$$

$$x(0) = ¼[x21(0) + x22(0)] = ¼[4] = \mathbf{1}$$
$$x(1) = ¼[x21(1) + j\,x22(1)] = ¼[2 + jj2] = \mathbf{0}$$
$$x(2) = ¼[x21(0) - x22(0)] = ¼[4] = \mathbf{1}$$
$$x(3) = ¼[x21(1) - j\,x22(1)] = ¼[2 - jj2] = \mathbf{1}$$

∴ x(n) = **{1, 0, 1, 1}**, the original data sequence.

Check by the IDFT (as Example 2.2):

$$x(nT) = F_D^{-1}[X(k)] = (1/N)\,\Sigma_{k=0}^{N-1}\,X(k)\,e^{jk\Omega nT} = (1/N)\,\Sigma_{k=0}^{N-1}\,X(k)\,e^{j2\pi kn/N} \quad (2.7)$$

With n = 0:

$$x(nT) = x(0) = (1/N)\,\Sigma_{k=0}^{3}\,X(k)\,e^{j2\pi k(0)/4}$$
$$= ¼\,[3 + j + 1 + (-j)]$$
$$= \mathbf{1}, \text{ as expected.}$$

With n = 1:

$$x(nT) = x(T) = (1/N)\,\Sigma_{k=0}^{3}\,X(k)\,e^{j2\pi k(1)/4}$$
$$= ¼\,[3 + j\,e^{j\pi/2} + 1\,e^{j\pi} + (-j)\,e^{j3\pi/2}]$$
$$= ¼\,[3 + j(j) + 1(-1) + (-j)(-j)]$$
$$= ¼\,[3 + (-1) + (-1) + (-1)]$$
$$= \mathbf{0}, \text{ as expected.}$$

With n = 2:

$$x(nT) = x(2T) = (1/N)\,\Sigma_{k=0}^{3}\,X(k)\,e^{j2\pi k(2)/4}$$
$$= ¼\,[3 + j\,e^{j\pi} + 1\,e^{j2\pi} + (-j)\,e^{j3\pi}]$$
$$= ¼\,[3 + j(-1) + 1(1) + (-j)(-1)]$$
$$= ¼\,[3 + (-j) + 1 + j]$$
$$= \mathbf{1}, \text{ as expected.}$$

With n = 3:

$$x(nT) = x(3T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(3)/4}$$
$$= \tfrac{1}{4}\ [3 + j\ e^{j3\pi/2} + 1\ e^{j3\pi} + (-j)\ e^{j9\pi/2}]$$
$$= \tfrac{1}{4}\ [3 + j(-j) + 1(-1) + (-j)(j)]$$
$$= \tfrac{1}{4}\ [3 + 1 + (-1) + 1]$$
$$= \mathbf{1}, \text{ as expected.}$$

∴ x(n) = **{1, 0, 1, 1}**, the original sequence.

(iii) Parseval's theorem:

$$\Sigma_{n=0}^{N-1}\ x^2(n) = (1/N)\ \Sigma_{k=0}^{N-1}\ |X(k)|^2 \quad (2.12)$$

|X(k)| = {3, 1, 1, 1}

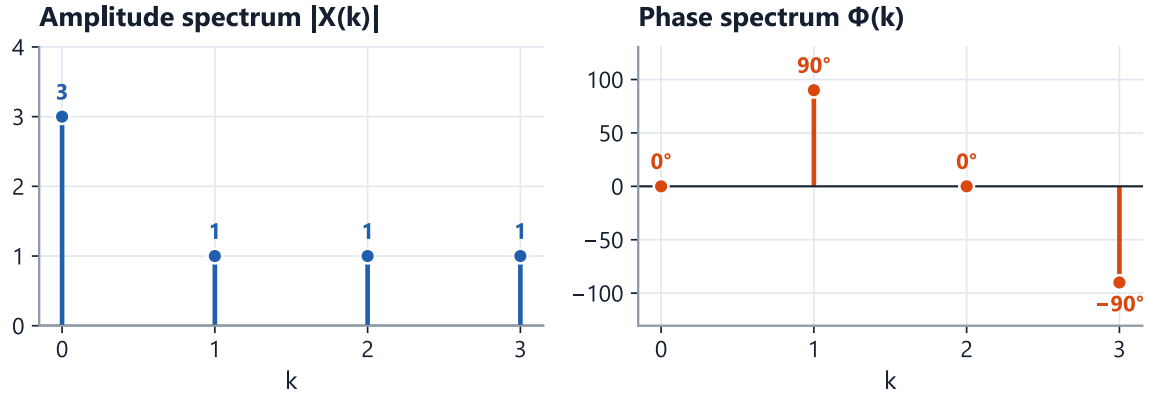
$$\text{L.H.S.} = 1^2 + 0^2 + 1^2 + 1^2 = \mathbf{3}$$
$$\text{R.H.S.} = (1/4)[3^2 + 1^2 + 1^2 + 1^2] = (1/4)(12) = \mathbf{3}$$

L.H.S. = R.H.S., so Parseval's theorem is verified.

(iv) Amplitude and phase spectra:

k	0	1	2	3
X(k)	3	j	1	−j
|X(k)|	3	1	1	1
Φ(k)	0°	90°	0°	−90°

Spectra of {1, 0, 1, 1}



(v) Computational saving (N = 4):

Operation	DFT	FFT	Saving
No. of complex multiplications	$N^2 = 16$	$(N/2)\ \log_2 N = 4$	**12**
No. of complex additions	$N(N-1) = 12$	$N\ \log_2 N = 8$	**4**

Final 18-1-2024 (Q2-c) Final 17-8-2026 (Q2-b) Final 23-1-2023 (Q3-b) Final 20-1-2025 (Q4-c) جه في

Midterm 8-12-2020 (Q2-c) | Midterm 31-7-2023 (Q2-b) | Final 2-1-2017 (Q2-c) | Final 12-1-2019 (Q2-c) | Book Assignment A2.2 | Midterm 2 (Q2-c)

P10 x(n) = {1, 0, 1, 0}

متكرر في الامتحانات

For the sequence x(n) = {1, 0, 1, 0}: (i) find the frequency components X(k) using the DFT (or FFT); (ii) check by calculating x(n) using the IFFT (or IDFT); (iii) verify Parseval's theorem; (iv) plot the amplitude and phase spectra; (v) find the computational saving of the FFT.

SOLUTION

(i) DFT method (as Example 2.1):

$$X(k) = \Sigma_{n=0}^{N-1} x(nT)\, e^{-jk\Omega nT} = \Sigma_{n=0}^{N-1} x(nT)\, e^{-j2\pi kn/N}, \quad \Omega = 2\pi/NT, \quad N = 4 \ (2.4)$$

x(0) = 1, x(T) = 0, x(2T) = 1, x(3T) = 0. It is required to find X(k) for k = 0, 1, 2, 3.

With k = 0:

$$\begin{aligned} X(0) &= \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(0)n/4} \\ &= x(0) + x(T) + x(2T) + x(3T) \\ &= 1 + 0 + 1 + 0 \\ &= \mathbf{2} \end{aligned}$$

Thus X(0) = 2, of magnitude 2 and phase angle Φ(0) = 0°.

With k = 1:

$$\begin{aligned} X(1) &= \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(1)n/4} \\ &= 1 + 0 + e^{-j\pi} + 0 \\ &= 1 + [\cos(\pi) - j\sin(\pi)] \\ &= 1 + (-1) \\ &= \mathbf{0} \end{aligned}$$

Thus X(1) = 0, of magnitude 0 and phase angle Φ(1) = 0°.

With k = 2:

$$\begin{aligned} X(2) &= \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(2)n/4} \\ &= 1 + 0 + e^{-j2\pi} + 0 \\ &= 1 + [\cos(2\pi) - j\sin(2\pi)] \\ &= 1 + 1 \\ &= \mathbf{2} \end{aligned}$$

Thus X(2) = 2, of magnitude 2 and phase angle Φ(2) = 0°.

With k = 3:

$$X(3) = \Sigma_{n=0}^{3}\, x(nT)\, e^{-j2\pi(3)n/4}$$
$$= 1 + 0 + e^{-j3\pi} + 0$$
$$= 1 + [\cos(3\pi) - j\sin(3\pi)]$$
$$= 1 + (-1)$$
$$= \mathbf{0}$$

Thus X(3) = 0, of magnitude 0 and phase angle Φ(3) = 0°.

So, the time series x(n) = {1, 0, 1, 0} has the DFT given by **X(k) = {2, 0, 2, 0}**.

FFT algorithm (as Example 2.3):

This is a four-point DFT with x0 = 1, x1 = 0, x2 = 1, x3 = 0, and

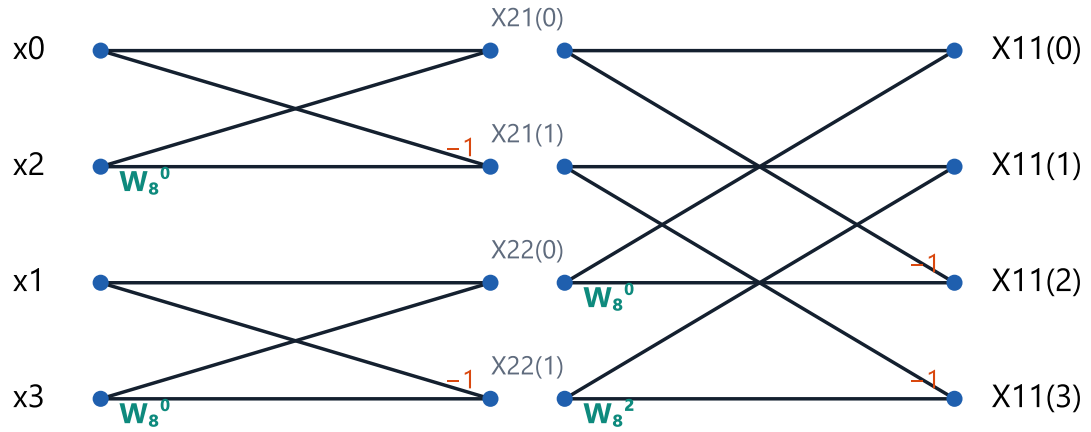
$$X1(k) = X11(k) + W_N^{\,k}\, X12(k), \quad k = 0, 1, 2, 3 \quad (2.24)$$

The re-ordered sequence is x0, x2, x1, x3. Using the top left-hand corner of Fig. 2.3 ($W_8^{\,0} = 1$, $W_8^{\,2} = e^{-j\pi/2} = -j$):

$$X21(0) = x0 + x2 = 1 + 1 = 2$$
$$X21(1) = x0 - x2 = 1 - 1 = 0$$
$$X22(0) = x1 + x3 = 0 + 0 = 0$$
$$X22(1) = x1 - x3 = 0 - 0 = 0$$

$$X11(0) = X21(0) + W_8^{0}\, X22(0) = 2 + 0 = \mathbf{2}$$
$$X11(1) = X21(1) + W_8^{2}\, X22(1) = 0 + e^{-j\pi/2}(0) = 0 - j(0) = \mathbf{0}$$
$$X11(2) = X21(0) - W_8^{0}\, X22(0) = 2 - 0 = \mathbf{2}$$
$$X11(3) = X21(1) - W_8^{2}\, X22(1) = 0 - e^{-j\pi/2}(0) = 0 + j(0) = \mathbf{0}$$

4-point DIT FFT (top left-hand corner of Fig. 2.3)



These values are the same as those obtained by the DFT.

(ii) Check by the IFFT (Sec. 2.6):

The IFFT (2.27) differs from the FFT (2.26) only in the factor 1/N and the sign of the exponent, so the same flowgraph is used with the re-ordered input X(0), X(2), X(1), X(3), $W_8^{-2} = e^{+j\pi/2} = +j$, and the result is divided by N = 4:

$$x21(0) = X(0) + X(2) = 4 \quad x21(1) = X(0) - X(2) = 0$$
$$x22(0) = X(1) + X(3) = 0 \quad x22(1) = X(1) - X(3) = 0$$

$$x(0) = \tfrac{1}{4}[x21(0) + x22(0)] = \tfrac{1}{4}[4] = \mathbf{1}$$
$$x(1) = \tfrac{1}{4}[x21(1) + j\,x22(1)] = \tfrac{1}{4}[0 + j0] = \mathbf{0}$$
$$x(2) = \tfrac{1}{4}[x21(0) - x22(0)] = \tfrac{1}{4}[4] = \mathbf{1}$$
$$x(3) = \tfrac{1}{4}[x21(1) - j\,x22(1)] = \tfrac{1}{4}[0 - j0] = \mathbf{0}$$

∴ x(n) = **{1, 0, 1, 0}**, the original data sequence.

Check by the IDFT (as Example 2.2):

$$x(nT) = F_D^{-1}[X(k)] = (1/N)\,\Sigma_{k=0}^{N-1}\,X(k)\,e^{jk\Omega nT} = (1/N)\,\Sigma_{k=0}^{N-1}\,X(k)\,e^{j2\pi kn/N} \quad (2.7)$$

With n = 0:

$$x(nT) = x(0) = (1/N)\,\Sigma_{k=0}^{3}\,X(k)\,e^{j2\pi k(0)/4}$$
$$= \tfrac{1}{4}\,[2 + 0 + 2 + 0]$$
$$= \mathbf{1}, \text{ as expected.}$$

With n = 1:

$$x(nT) = x(T) = (1/N)\,\Sigma_{k=0}^{3}\,X(k)\,e^{j2\pi k(1)/4}$$
$$= \tfrac{1}{4}\,[2 + 0 + 2\,e^{j\pi} + 0]$$
$$= \tfrac{1}{4}\,[2 + 2(-1)]$$
$$= \tfrac{1}{4}\,[2 + (-2)]$$
$$= \mathbf{0}, \text{ as expected.}$$

With n = 2:

$$x(nT) = x(2T) = (1/N)\,\Sigma_{k=0}^{3}\,X(k)\,e^{j2\pi k(2)/4}$$
$$= \tfrac{1}{4}\,[2 + 0 + 2\,e^{j2\pi} + 0]$$
$$= \tfrac{1}{4}\,[2 + 2(1)]$$
$$= \tfrac{1}{4}\,[2 + 2]$$
$$= \mathbf{1}, \text{ as expected.}$$

With n = 3:

$$x(nT) = x(3T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(3)/4}$$
$$= \tfrac{1}{4}\ [2 + 0 + 2\ e^{j3\pi} + 0]$$
$$= \tfrac{1}{4}\ [2 + 2(-1)]$$
$$= \tfrac{1}{4}\ [2 + (-2)]$$
$$= \mathbf{0}, \text{ as expected.}$$

∴ x(n) = **{1, 0, 1, 0}**, the original sequence.

(iii) Parseval's theorem:

$$\Sigma_{n=0}^{N-1}\ x^2(n) = (1/N)\ \Sigma_{k=0}^{N-1}\ |X(k)|^2 \quad (2.12)$$

|X(k)| = {2, 0, 2, 0}

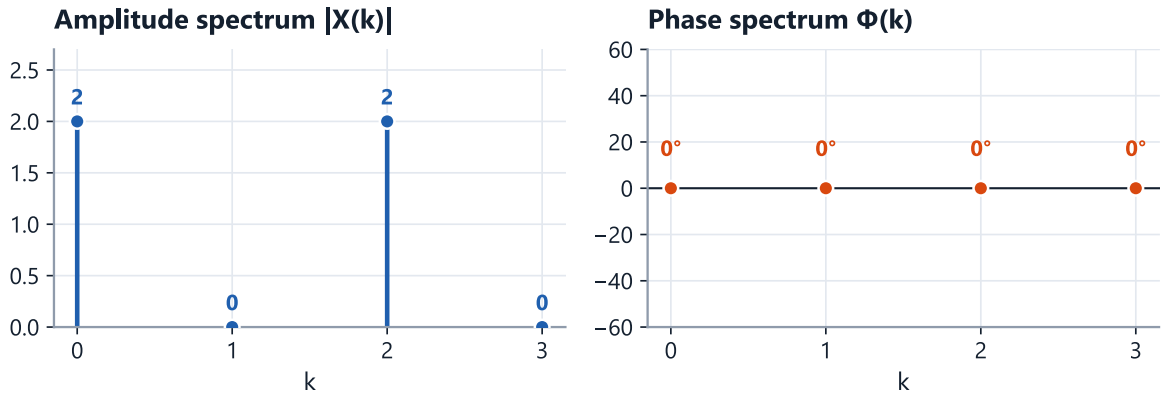
$$\text{L.H.S.} = 1^2 + 0^2 + 1^2 + 0^2 = \mathbf{2}$$
$$\text{R.H.S.} = (1/4)[2^2 + 0^2 + 2^2 + 0^2] = (1/4)(8) = \mathbf{2}$$

L.H.S. = R.H.S., so Parseval's theorem is verified.

(iv) Amplitude and phase spectra:

k	0	1	2	3
X(k)	2	0	2	0
|X(k)|	2	0	2	0
Φ(k)	0°	0°	0°	0°

Spectra of {1, 0, 1, 0}



(v) Computational saving (N = 4):

Operation	DFT	FFT	Saving
No. of complex multiplications	$N^2 = 16$	$(N/2)\ \log_2 N = 4$	**12**
No. of complex additions	$N(N-1) = 12$	$N\ \log_2 N = 8$	**4**

Midterm 12-11-2025 (Q2-b) Final 7-6-2023 (Q1-c) Final 11-6-2023 (Q1-c) جه في

Midterm 28-11-2021 (Q2-b)

P11 x(n) = {1, 1, 0, 1}

For the sequence x(n) = {1, 1, 0, 1}: (i) find the frequency components X(k) using the DFT (or FFT); (ii) check by calculating x(n) using the IFFT (or IDFT); (iii) verify Parseval's theorem; (iv) plot the amplitude and phase spectra; (v) find the computational saving of the FFT.

SOLUTION

(i) DFT method (as Example 2.1):

$$X(k) = \Sigma_{n=0}^{N-1} x(nT)\, e^{-jk\Omega nT} = \Sigma_{n=0}^{N-1} x(nT)\, e^{-j2\pi kn/N}, \quad \Omega = 2\pi/NT, \quad N = 4 \ (2.4)$$

x(0) = 1, x(T) = 1, x(2T) = 0, x(3T) = 1. It is required to find X(k) for k = 0, 1, 2, 3.

With k = 0:

$$\begin{aligned} X(0) &= \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(0)n/4} \\ &= x(0) + x(T) + x(2T) + x(3T) \\ &= 1 + 1 + 0 + 1 \\ &= \mathbf{3} \end{aligned}$$

Thus X(0) = 3, of magnitude 3 and phase angle Φ(0) = 0°.

With k = 1:

$$\begin{aligned} X(1) &= \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(1)n/4} \\ &= 1 + e^{-j\pi/2} + 0 + e^{-j3\pi/2} \\ &= 1 + [\cos(\pi/2) - j\sin(\pi/2)] + [\cos(3\pi/2) - j\sin(3\pi/2)] \\ &= 1 + (-j) + j \\ &= \mathbf{1} \end{aligned}$$

Thus X(1) = 1, of magnitude 1 and phase angle Φ(1) = 0°.

With k = 2:

$$\begin{aligned} X(2) &= \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(2)n/4} \\ &= 1 + e^{-j\pi} + 0 + e^{-j3\pi} \\ &= 1 + [\cos(\pi) - j\sin(\pi)] + [\cos(3\pi) - j\sin(3\pi)] \\ &= 1 + (-1) + (-1) \\ &= \mathbf{-1} \end{aligned}$$

Thus X(2) = −1, of magnitude 1 and phase angle Φ(2) = 180°.

With k = 3:

$$X(3) = \Sigma_{n=0}^{3}\, x(nT)\, e^{-j2\pi(3)n/4}$$
$$= 1 + e^{-j3\pi/2} + 0 + e^{-j9\pi/2}$$
$$= 1 + [\cos(3\pi/2) - j\sin(3\pi/2)] + [\cos(9\pi/2) - j\sin(9\pi/2)]$$
$$= 1 + j + (-j)$$
$$= \mathbf{1}$$

Thus X(3) = 1, of magnitude 1 and phase angle Φ(3) = 0°.

So, the time series x(n) = {1, 1, 0, 1} has the DFT given by **X(k) = {3, 1, −1, 1}**.

FFT algorithm (as Example 2.3):

This is a four-point DFT with x0 = 1, x1 = 1, x2 = 0, x3 = 1, and

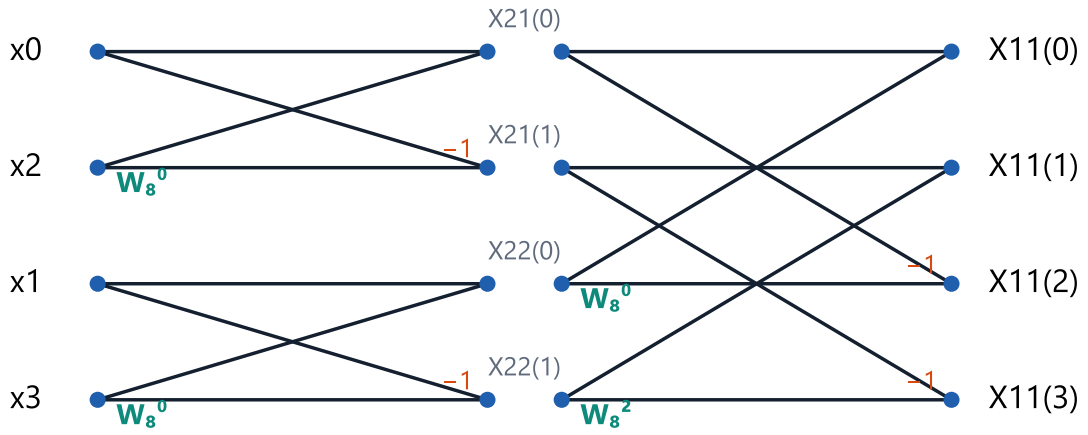
$$X1(k) = X11(k) + W_N^{k}\, X12(k), \quad k = 0, 1, 2, 3 \quad (2.24)$$

The re-ordered sequence is x0, x2, x1, x3. Using the top left-hand corner of Fig. 2.3 ($W_8^0 = 1$, $W_8^2 = e^{-j\pi/2} = -j$):

$$X21(0) = x0 + x2 = 1 + 0 = 1$$
$$X21(1) = x0 - x2 = 1 - 0 = 1$$
$$X22(0) = x1 + x3 = 1 + 1 = 2$$
$$X22(1) = x1 - x3 = 1 - 1 = 0$$

$$X11(0) = X21(0) + W_8^0\, X22(0) = 1 + 2 = \mathbf{3}$$
$$X11(1) = X21(1) + W_8^2\, X22(1) = 1 + e^{-j\pi/2}(0) = 1 - j(0) = \mathbf{1}$$
$$X11(2) = X21(0) - W_8^0\, X22(0) = 1 - 2 = \mathbf{-1}$$
$$X11(3) = X21(1) - W_8^2\, X22(1) = 1 - e^{-j\pi/2}(0) = 1 + j(0) = \mathbf{1}$$

4-point DIT FFT (top left-hand corner of Fig. 2.3)



These values are the same as those obtained by the DFT.

(ii) Check by the IFFT (Sec. 2.6):

The IFFT (2.27) differs from the FFT (2.26) only in the factor 1/N and the sign of the exponent, so the same flowgraph is used with the re-ordered input X(0), X(2), X(1), X(3), $W_8^{-2} = e^{+j\pi/2} = +j$, and the result is divided by N = 4:

$$x21(0) = X(0) + X(2) = 2 \quad x21(1) = X(0) - X(2) = 4$$
$$x22(0) = X(1) + X(3) = 2 \quad x22(1) = X(1) - X(3) = 0$$

$$x(0) = \tfrac{1}{4}[x21(0) + x22(0)] = \tfrac{1}{4}[4] = \mathbf{1}$$
$$x(1) = \tfrac{1}{4}[x21(1) + j\,x22(1)] = \tfrac{1}{4}[4 + j0] = \mathbf{1}$$
$$x(2) = \tfrac{1}{4}[x21(0) - x22(0)] = \tfrac{1}{4}[0] = \mathbf{0}$$
$$x(3) = \tfrac{1}{4}[x21(1) - j\,x22(1)] = \tfrac{1}{4}[4 - j0] = \mathbf{1}$$

∴ x(n) = **{1, 1, 0, 1}**, the original data sequence.

Check by the IDFT (as Example 2.2):

$$x(nT) = F_D^{-1}[X(k)] = (1/N)\,\Sigma_{k=0}^{N-1}\,X(k)\,e^{jk\Omega nT} = (1/N)\,\Sigma_{k=0}^{N-1}\,X(k)\,e^{j2\pi kn/N} \quad (2.7)$$

With n = 0:

$$x(nT) = x(0) = (1/N)\,\Sigma_{k=0}^{3}\,X(k)\,e^{j2\pi k(0)/4}$$
$$= \tfrac{1}{4}\,[3 + 1 + (-1) + 1]$$
$$= \mathbf{1}, \text{ as expected.}$$

With n = 1:

$$x(nT) = x(T) = (1/N)\,\Sigma_{k=0}^{3}\,X(k)\,e^{j2\pi k(1)/4}$$
$$= \tfrac{1}{4}\,[3 + 1\,e^{j\pi/2} + (-1)\,e^{j\pi} + 1\,e^{j3\pi/2}]$$
$$= \tfrac{1}{4}\,[3 + 1(j) + (-1)(-1) + 1(-j)]$$
$$= \tfrac{1}{4}\,[3 + j + 1 + (-j)]$$
$$= \mathbf{1}, \text{ as expected.}$$

With n = 2:

$$x(nT) = x(2T) = (1/N)\,\Sigma_{k=0}^{3}\,X(k)\,e^{j2\pi k(2)/4}$$
$$= \tfrac{1}{4}\,[3 + 1\,e^{j\pi} + (-1)\,e^{j2\pi} + 1\,e^{j3\pi}]$$
$$= \tfrac{1}{4}\,[3 + 1(-1) + (-1)(1) + 1(-1)]$$
$$= \tfrac{1}{4}\,[3 + (-1) + (-1) + (-1)]$$
$$= \mathbf{0}, \text{ as expected.}$$

With n = 3:

$$x(nT) = x(3T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(3)/4}$$
$$= \tfrac{1}{4}\ [3 + 1\ e^{j3\pi/2} + (-1)\ e^{j3\pi} + 1\ e^{j9\pi/2}]$$
$$= \tfrac{1}{4}\ [3 + 1(-j) + (-1)(-1) + 1(j)]$$
$$= \tfrac{1}{4}\ [3 + (-j) + 1 + j]$$
$$= \mathbf{1}, \text{ as expected.}$$

∴ x(n) = **{1, 1, 0, 1}**, the original sequence.

(iii) Parseval's theorem:

$$\Sigma_{n=0}^{N-1}\ x^2(n) = (1/N)\ \Sigma_{k=0}^{N-1}\ |X(k)|^2 \quad (2.12)$$

|X(k)| = {3, 1, 1, 1}

$$\text{L.H.S.} = 1^2 + 1^2 + 0^2 + 1^2 = \mathbf{3}$$
$$\text{R.H.S.} = (1/4)[3^2 + 1^2 + 1^2 + 1^2] = (1/4)(12) = \mathbf{3}$$

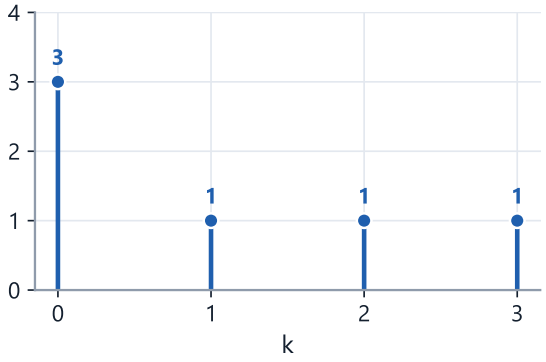
L.H.S. = R.H.S., so Parseval's theorem is verified.

(iv) Amplitude and phase spectra:

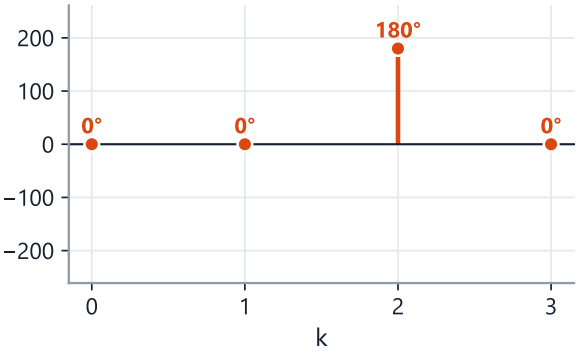
k	0	1	2	3
X(k)	3	1	−1	1
|X(k)|	3	1	1	1
Φ(k)	0°	0°	180°	0°

Spectra of {1, 1, 0, 1}

Amplitude spectrum |X(k)|



Phase spectrum Φ(k)



(v) Computational saving (N = 4):

Operation	DFT	FFT	Saving
No. of complex multiplications	$N^2 = 16$	$(N/2)\ \log_2 N = 4$	**12**
No. of complex additions	$N(N-1) = 12$	$N\ \log_2 N = 8$	**4**

Midterm 1 Model-2 (Q2-c) Final 9-6-2026 (Q2-b) جه في

P12 x(n) = {1, 0, 0, 1}

For the sequence x(n) = {1, 0, 0, 1}: (i) find the frequency components X(k) using the DFT (or FFT); (ii) check by calculating x(n) using the IFFT (or IDFT); (iii) verify Parseval's theorem; (iv) plot the amplitude and phase spectra; (v) find the computational saving of the FFT.

SOLUTION

(i) DFT method (as Example 2.1):

$$X(k) = \Sigma_{n=0}^{N-1} x(nT)\, e^{-jk\Omega nT} = \Sigma_{n=0}^{N-1} x(nT)\, e^{-j2\pi kn/N}, \quad \Omega = 2\pi/NT, \quad N = 4 \text{ (2.4)}$$

x(0) = 1, x(T) = 0, x(2T) = 0, x(3T) = 1. It is required to find X(k) for k = 0, 1, 2, 3.

With k = 0:

$$\begin{aligned} X(0) &= \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(0)n/4} \\ &= x(0) + x(T) + x(2T) + x(3T) \\ &= 1 + 0 + 0 + 1 \\ &= \mathbf{2} \end{aligned}$$

Thus X(0) = 2, of magnitude 2 and phase angle Φ(0) = 0°.

With k = 1:

$$\begin{aligned} X(1) &= \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(1)n/4} \\ &= 1 + 0 + 0 + e^{-j3\pi/2} \\ &= 1 + [\cos(3\pi/2) - j\sin(3\pi/2)] \\ &= 1 + j \\ &= \mathbf{1 + j} \end{aligned}$$

Thus X(1) = 1 + j, of magnitude √2 = 1.414 and phase angle Φ(1) = 45°.

With k = 2:

$$\begin{aligned} X(2) &= \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(2)n/4} \\ &= 1 + 0 + 0 + e^{-j3\pi} \\ &= 1 + [\cos(3\pi) - j\sin(3\pi)] \\ &= 1 + (-1) \\ &= \mathbf{0} \end{aligned}$$

Thus X(2) = 0, of magnitude 0 and phase angle Φ(2) = 0°.

With k = 3:

$$X(3) = \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(3)n/4}$$
$$= 1 + 0 + 0 + e^{-j9\pi/2}$$
$$= 1 + [\cos(9\pi/2) - j \sin(9\pi/2)]$$
$$= 1 + (-j)$$
$$= \mathbf{1 - j}$$

Thus X(3) = 1 − j, of magnitude √2 = 1.414 and phase angle Φ(3) = −45°.

So, the time series x(n) = {1, 0, 0, 1} has the DFT given by **X(k) = {2, 1 + j, 0, 1 − j}**.

FFT algorithm (as Example 2.3):

This is a four-point DFT with x0 = 1, x1 = 0, x2 = 0, x3 = 1, and

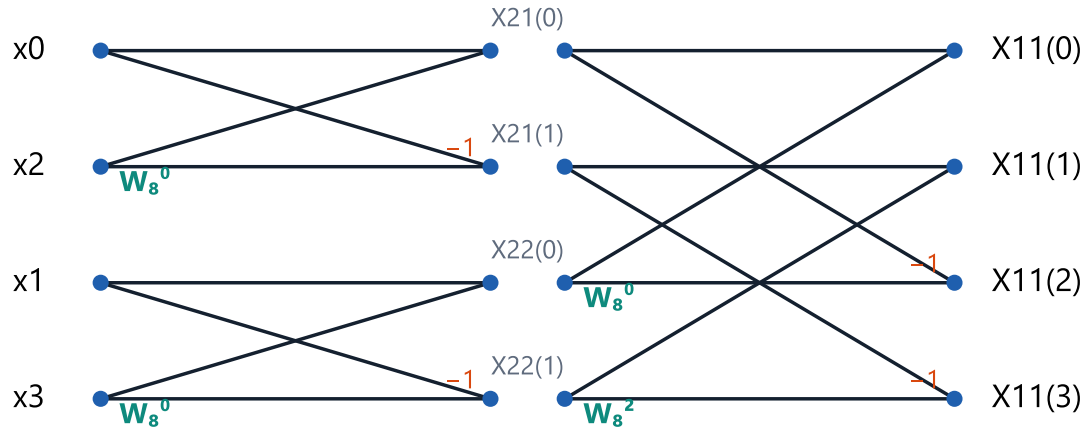
$$X1(k) = X11(k) + W_N^k\, X12(k), \quad k = 0, 1, 2, 3 \quad (2.24)$$

The re-ordered sequence is x0, x2, x1, x3. Using the top left-hand corner of Fig. 2.3 ($W_8^0 = 1$, $W_8^2 = e^{-j\pi/2} = -j$):

$$X21(0) = x0 + x2 = 1 + 0 = 1$$
$$X21(1) = x0 - x2 = 1 - 0 = 1$$
$$X22(0) = x1 + x3 = 0 + 1 = 1$$
$$X22(1) = x1 - x3 = 0 - 1 = -1$$

$$X11(0) = X21(0) + W_8^0\, X22(0) = 1 + 1 = \mathbf{2}$$
$$X11(1) = X21(1) + W_8^2\, X22(1) = 1 + e^{-j\pi/2}(-1) = 1 - j(-1) = \mathbf{1 + j}$$
$$X11(2) = X21(0) - W_8^0\, X22(0) = 1 - 1 = \mathbf{0}$$
$$X11(3) = X21(1) - W_8^2\, X22(1) = 1 - e^{-j\pi/2}(-1) = 1 + j(-1) = \mathbf{1 - j}$$

4-point DIT FFT (top left-hand corner of Fig. 2.3)



These values are the same as those obtained by the DFT.

(ii) Check by the IFFT (Sec. 2.6):

The IFFT (2.27) differs from the FFT (2.26) only in the factor 1/N and the sign of the exponent, so the same flowgraph is used with the re-ordered input X(0), X(2), X(1), X(3), $W_8^{-2} = e^{+j\pi/2} = +j$, and the result is divided by N = 4:

$$x21(0) = X(0) + X(2) = 2 \quad x21(1) = X(0) - X(2) = 2$$
$$x22(0) = X(1) + X(3) = 2 \quad x22(1) = X(1) - X(3) = j2$$

$$x(0) = ¼[x21(0) + x22(0)] = ¼[4] = \mathbf{1}$$
$$x(1) = ¼[x21(1) + j\,x22(1)] = ¼[2 + jj2] = \mathbf{0}$$
$$x(2) = ¼[x21(0) - x22(0)] = ¼[0] = \mathbf{0}$$
$$x(3) = ¼[x21(1) - j\,x22(1)] = ¼[2 - jj2] = \mathbf{1}$$

∴ x(n) = **{1, 0, 0, 1}**, the original data sequence.

Check by the IDFT (as Example 2.2):

$$x(nT) = F_D^{-1}[X(k)] = (1/N)\,\Sigma_{k=0}^{N-1}\,X(k)\,e^{jk\Omega nT} = (1/N)\,\Sigma_{k=0}^{N-1}\,X(k)\,e^{j2\pi kn/N} \quad (2.7)$$

With n = 0:

$$x(nT) = x(0) = (1/N)\,\Sigma_{k=0}^{3}\,X(k)\,e^{j2\pi k(0)/4}$$
$$= ¼\,[2 + (1 + j) + 0 + (1 - j)]$$
$$= \mathbf{1}, \text{ as expected.}$$

With n = 1:

$$x(nT) = x(T) = (1/N)\,\Sigma_{k=0}^{3}\,X(k)\,e^{j2\pi k(1)/4}$$
$$= ¼\,[2 + (1 + j)\,e^{j\pi/2} + 0 + (1 - j)\,e^{j3\pi/2}]$$
$$= ¼\,[2 + (1 + j)(j) + (1 - j)(-j)]$$
$$= ¼\,[2 + (-1 + j) + (-1 - j)]$$
$$= \mathbf{0}, \text{ as expected.}$$

With n = 2:

$$x(nT) = x(2T) = (1/N)\,\Sigma_{k=0}^{3}\,X(k)\,e^{j2\pi k(2)/4}$$
$$= ¼\,[2 + (1 + j)\,e^{j\pi} + 0 + (1 - j)\,e^{j3\pi}]$$
$$= ¼\,[2 + (1 + j)(-1) + (1 - j)(-1)]$$
$$= ¼\,[2 + (-1 - j) + (-1 + j)]$$
$$= \mathbf{0}, \text{ as expected.}$$

With n = 3:

$$x(nT) = x(3T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(3)/4}$$

$$= \tfrac{1}{4}\ [2 + (1 + j)\ e^{j3\pi/2} + 0 + (1 - j)\ e^{j9\pi/2}]$$

$$= \tfrac{1}{4}\ [2 + (1 + j)(-j) + (1 - j)(j)]$$

$$= \tfrac{1}{4}\ [2 + (1 - j) + (1 + j)]$$

$$= \mathbf{1}, \text{ as expected.}$$

∴ x(n) = **{1, 0, 0, 1}**, the original sequence.

(iii) Parseval's theorem:

$$\Sigma_{n=0}^{N-1}\ x^2(n) = (1/N)\ \Sigma_{k=0}^{N-1}\ |X(k)|^2 \quad (2.12)$$

|X(k)| = {2, √2, 0, √2}

$$\text{L.H.S.} = 1^2 + 0^2 + 0^2 + 1^2 = \mathbf{2}$$

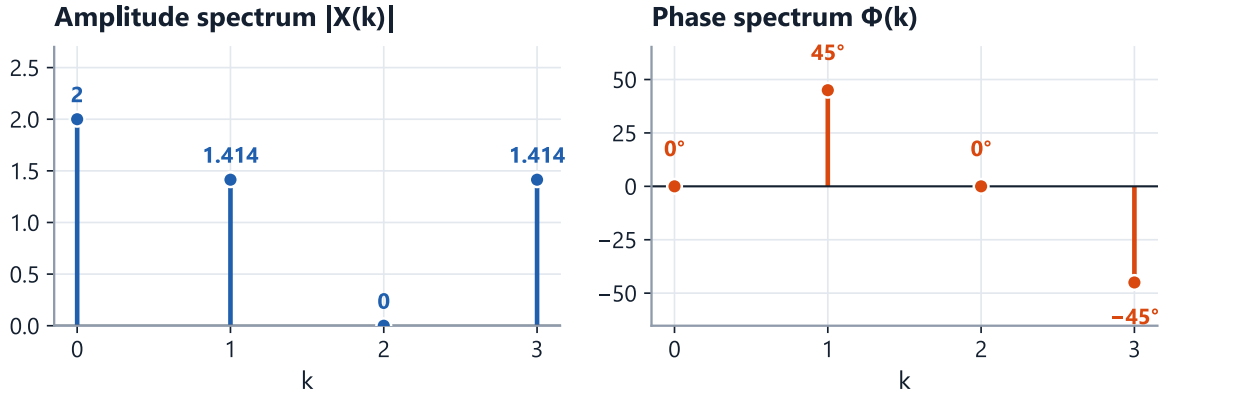
$$\text{R.H.S.} = (1/4)[2^2 + \sqrt{2}^2 + 0^2 + \sqrt{2}^2] = (1/4)(8) = \mathbf{2}$$

L.H.S. = R.H.S., so Parseval's theorem is verified.

(iv) Amplitude and phase spectra:

k	0	1	2	3
X(k)	2	1 + j	0	1 − j
|X(k)|	2	√2 = 1.414	0	√2 = 1.414
Φ(k)	0°	45°	0°	−45°

Spectra of {1, 0, 0, 1}



(v) Computational saving (N = 4):

Operation	DFT	FFT	Saving
No. of complex multiplications	$N^2 = 16$	$(N/2)\ \log_2 N = 4$	**12**
No. of complex additions	$N(N - 1) = 12$	$N\ \log_2 N = 8$	**4**

جه في Final 23-12-2017 (Q2-c) Final 28-12-2019 (Q2-b) Midterm 26-7-2026 (Q2-b)